

3rd Applied Active Inference Symposium ~ "Enacting Ecosystems of Shared Intelligence"

~ 2nd session | 6:33:15

Hello and welcome everyone. Welcome back. This is the second interval of the third Applied Active Inference Symposium at the Active Inference Institute on August 22nd, 2023. This is going to be another packed and exciting interval and we're kicking it off with Jean-Francois Cloutier, a collective of theorizers, First Steps. So JF, thank you for joining and to you for this presentation. And if anybody has questions for this presentation or any of the others, please just put it in the live chat and I'll do it. I can. So thanks JF to you.

Well, thank you, Daniel. I'm a software engineer at a company called Smart Rent, where I write software for smart home systems.

I also work on a research project at the Active Inference Institute called the Robotics and Embodied Project. My current focus is on supervised learning in active inference agents. In my presentation today, which is titled A Collective of Theorizers First Steps, I'll first start with a brief recap of the project, then dive into recent progress, and then I'll conclude with what I see as the next steps of the project.

Well, since 2017, I've been experimenting with LEGO robots and models of cognition. I do this for my own education

because, like I think most of us, I understand something better when I build it. What you see here is my latest robot model. It's a rover. It has a whole bunch of sensors and a number of effectors, actuators. As a matter of fact,

those sensors and actuators can be understood as forming the Markov blanket of my robot.

Last year, I presented at the symposium about the history of the project, its then status, and its ambitions.

I presented a cognitive model that implemented predictive processing from an active inference perspective. So there were generative models, there was predictions, there was prediction errors, the models, and these generative models animated the robots so that they would roam, avoid obstacles, observe their companion, and also in a very simplistic way build a theory of mind of the other robot.

They could, from observations, infer that the other robot had detected food, food being presented here as a sheet of paper on the floor, and then would kind of try to track the other robot to also get to the food.

And fun stuff like this. The implementation combined multiprocessing and functional computing. There was nothing probabilistic in the implementation.

I thought maybe we would see them in action. That's what was presented last year, but just to get a sense of what they do. I have two robots here, one named Carl and Andy. I named them before I knew I was going to present at the conference. So they benefited from a number of training sessions. They learned how to find which policies would achieve their goals better than others.

And one here on the left found the food rather straight right away, but got closer to the pedestal where the beacon is that simulates the scent of food.

The fear was getting into a collision and then backs off in a hurry, as we'll see.

The other robot observes all this, sees the other, the robot backing off as being in a panic and

decides to share in this emotion and backs up as well. So, you know, fun stuff.

The, since from the beginning of the M implemented different cognition models, but they all have in common the fact that they implement a society of mind. So what is a society of mind? A society of mind is a concept by which the mind is not a monolithic structure, but a composition of simple actors, independent actors that interact with each other in simple ways.

That view of the mind was put forth by Marvin Minsky 50 years ago. So this is not recent.

Again, what I presented last year was a society of mind containing a hierarchy of,

I call them cognition actors. Each cognition actor is an independent process. Each one has a scope, a level of abstraction as well. So for example, you would have a cognition actor that's concerned with the location of food. And it would have beliefs and perceptions about food location. And these would feed into a higher level cognition actor, let's say, food approach, which is concerned about getting closer to the food, and so forth and so on. These function actors communicate, again, in simple ways, this is a society of mind. And they communicate by emitting predictions, predictions about the beliefs of other cognition actors. And they communicate by emitting prediction errors when predictions made about their

beliefs by others are inaccurate. That leads to cognition actors processing these prediction errors and combining them with their own predictions to create an updated set of perceptions that are synthesized into

beliefs. And these beliefs lead to actions in order to eliminate negative beliefs and validate positive beliefs.

Now, it's from the interactions of all these cognition actors that seemingly purposeful behaviors emerge.

So that was successful. But learning was very limited. This hierarchy of cognition actors was given, was predefined.

And the only learning that a robot would do was to discover which action policies tended to be more effective.

So clearly, my robots are not monolithic active inference agents. And so the question is, why would I use a society of mind architecture? Well, because I subscribe to the notion that all intelligence is used to embrace collective intelligence. And this paper makes the argument quite cogently.

And I'm going to cite a number of papers which were important to the evolution of my thinking. This is one of them.

This paper sees intelligence as a process, not a property.

It's a process enacted by the interacting parts, as opposed to, again, a property of individuals.

So, a society of mind is basically that.

It's actually a number of interacting processes that together, from which emerge apparently intelligent behaviors.

Now, my own personal current definition of intelligence is self-sustaining and active sensemaking, since making is really important, by autonomous agent system in a dissipative and dynamic environment. And this guides all the work I do in this project.

Now, if you want a description, a detailed description of where my project stood a year ago, there's a paper that was published, that I published on ZenoDome, which you can look up.

Now, over the last year, I wanted to move away from a pre-built, a given society of mind,

and toward a learned society of mind.

I wanted to see if I could program an autonomous robot to evolve its society of mind through its interactions with its environment.

Now, one might think that programming autonomy is an oxymoron.

It would be a good point, but I don't think it is.

If the program I write and install my robot imparts constitutive autonomy,

which is enable the robot to constitute its own identity,

and if it imparts adaptivity, which enables the robot to modify itself from its interactions with the environment, then I think that the robot will be truly autonomous.

Now, for autonomy to exist inherently in the robot,

something in the robot must be at stake.

And in this case, it's the survival of the robot's society of mind.

Now, the physical structure of the robot is not at stake.

Its survival is not at stake, unless, of course, it falls off a shelf.

But as I intend to have my robot develop its society of mind from experiences,

I also expect that if it fails, that the society of mind will perish in its attempt to grow and sustain itself.

So that's what's at stake in this current, going forward in this project,

is the survival of the robot's society of mind.

And this survival would be an expression of being by doing.

The robot will need to act and interact in its environment in order to survive.

Now, whatever sense it's going to make of its environment will then be grounded in this survival imperative.

In essence, the robot's society of mind will have skin in the game.

If this weren't the case, then sense-making would actually reside somewhere else.

It would reside in the mind of the programmer, in my mind, as I observed the robots.

But I would like the sense-making to be grounded in the survival of the robot's mind itself.

So that's key.

Now, what do I mean by an evolving and growing society of mind?

So instead of the given society of mind, which I showed earlier,

I want the cognition actors to be created to connect with each other dynamically through interactions with the environment.

So I want an active, self-organizing, self-optimizing collective of cognition actors.

Now, is this feasible?

Well, that's a big question.

In order to, in trying to answer this question,

I'll be also answering questions like,

what must be given a priori and what can be discovered?

Well, I already have elements of an answer.

I know that the sensors and the effectors and the primitive cognition actors that wrap them will be given.

This will be like what you're born with, basically.

And there might be a metacognition actor,
which role will be to oversee and guide the evolution of all other cognition actors.
But that's my hypothesis.

I can imagine a test environment for my robot that will test its ability to survive.

I can imagine, for example, that as the robot moves around or computes,
it uses a limited store of energy.

It's simulated.

And that store of energy is replenished when the robot consumes food.

And that would be by being on top of pieces of paper, colored pieces of paper on the floor.

And I'm going to make sure that it needs two sources of food in order to survive.

That would be represented by a yellow paper and a green paper, for example.

So that to prevent the robot from just simply finding one source of food and just stationing itself over it.

So the environment will also contain obstacles.

So the robot will need to learn how to navigate, avoid obstacles,
locate different sources of food,
get to them,

and make sure that it alternate between various sources of food in order to survive.

And the society of mind that will evolve will hopefully evolve to successfully do this.

Else, if it doesn't,
then it will shrink as resources disappear and essentially die.

So that's what's at stake for this robot.

Now, this effort employs a number of frameworks.

And by framework, I mean a useful system of concept and constraints that guide the implementation.

Well, there's obviously the free energy principle and the active inference framework.

However, I see this, I see active inference as an as-if framework.

It describes the what.

What must be achieved.

In this case, reduction of the agent's variational free energy.

But it doesn't guide me as to the implementation, how to build the robot.

For this, I need as-is frameworks.

And I'll be using two frameworks.

One, which I've been using since the beginning of the project, which is the actor model.

The actor model views computation as a diversity of processes.

Processes that are independent, have their own private internal state,
and who communicate with one another strictly through messages.

Yesterday in Interpol 1, Keith Duggar presented on the actor model
and made the case that we should use the actor model to implement active inference agents.

Well, I wholeheartedly agree with him.

The other framework that I'm going to be using is a special case of symbolic AI called the Apperception Engine.

And much of the presentation will be about the Apperception Engine and its implementation.

So here's where we are.

The project is located at the intersection of active inference as a domain, the what.

And society of mind as an architecture and symbolic AI as a form of computing.

So that's where the project is at this intersection.

Now, so where to begin?

So I want to dive into a more extensive form of learning.

And the first step, logically, is to learn how to predict.

So I want to enable a single cognition actor.

We'll start with a single cognition actor to learn how to make sense of its local environment, its so-called umwelt.

And making sense implies at a minimum to be able to predict incoming sensation.

So it needs to learn to predict.

So what will be given to a cognition actor?

Well, there will be a history of sensations broken into discrete units of time.

So time n minus 3, n minus 2, n minus 1, time n , which is the present moment.

So these would be remembered observations.

And then what we want to get out of this is the ability to predict the next incoming set of sensations at time t equals n plus 1, n plus 2.

And for this, to be able to predict future observations, we need some kind of predictor function that is learned from the remembered observation.

Now, this predictive capability can be built in two very general ways.

One is from statistics.

One is from statistics.

So doing pattern analysis and being able to predict what's most probable, which is the standard current machine learning approach.

Or we could predict from an understanding of the observations by developing a causal model of what produced these sensations.

And from this understanding, predict what rationally should be observed next.

So this is all about sensemaking.

And now what is sensemaking?

How do I understand sensemaking?

Well, to rationally predict incoming sensory inputs, one must make sense of them.

That's what making sense means to me.

And to make sense of sensory inputs means to derive meaningful experiences from them.

Okay, so I'm going to try to give you the postage stamp version of Immanuel Kant's theory,

focusing on his synthetic unity of apperception. Well, first of all, there's the real world, which is outside of our direct experience. It's the noumena, it's forever hidden from us, as in as is reality. But we do, it impinges on our sensorium. And then so we have a number of intuitions, sight, sound, touch, smell, that are initially separate. And then we need to network them, connect them both in time and in space. In space, is the sound and the sight describing a single thing? Is one thing behind or inside another? And in time, is this happening before, after something else? And then at a higher level, meaning is given to these networked intuitions, sensations, via concepts, and judgments, rules, which are generalizations as to what can and cannot be. And this is what we experience.

Now, it turns out that a synthetic unity of apperception is a blueprint for automating sensemaking. And it's kind of interesting, I think, that 18th century philosophy would be relevant to 21st century technology development. And this is what happened and was published by Richard Evans and Aal in the paper,

Making Sense of Sensory Input, where they developed the apperception engine. They took Kant's synthetic unity of apperception as software requirements and successfully implemented them into a piece of software, the apperception engine, and applied it to a number of exercises where they got a very good result. So the apperception engine is an instance of machine learning, it's unsupervised machine learning, and it operates on very small data sets, and generates human readable generative models. And when I read the paper, I realized, well, that's exactly what my robot is needed. So what does a apperception engine do? Well, given a sequence of observed states, it finds a generative model that can recreate past states, but most importantly, predict future states. And the state is defined as a set of simultaneous observations, sensations, intuitions.

So an apperception engine searches for a causal theory that can recreate observations. I say searches, because this causal theory is not determined by the observations. It has to be found, it has to be discovered. But once it is found, then it can be validated against the observations and see if it can recreate them and augment them into the future as well as into the past. So what is a causal theory? A causal

theory is a logic program that has a number of components. In the causal theory, there will be the objects and the predicates from the observed relations. So from the observations, we can extract what objects were observed, and what properties of these objects were observed, maybe what relationships of these objects were observed. That's the start. Then we have latent object types, objects and predicates. So we may want to imagine, causal theory imagines hidden objects, maybe in types of objects, and maybe hidden properties and relationships between objects, latent meaning unobserved. And given both the observed and unobserved objects and predicates, it derives rules. First of all, constraints on those predicates, what's permissible. So for example, being in front of cannot, A cannot be in front of B, and at the same time behind B. So an object cannot be in front of another and behind it. So there are constraints on predicates. And then there are rules that apply to any simultaneous sets of observations, but what they must conform to. Then there are rules that, given a state, will infer the next state.

And then maybe some initial states from which we can run the causal theory.

So what makes a causal theory unified? Well, first of all, it needs to be unified in order to make sense of the observation. There are various dimensions. So if a causal theory involves a number of objects, all these objects must be directly or indirectly related. There's no object that just floats in space, totally independent of the other objects. So they must all be related. So they're spatially unified.

All predicates that make up the causal theory, like on, off, behind, and in front, they must be constrained. So that, for example, in front cannot be at the same time as behind, or that a light cannot be both turned on and turned off. So there's some restrictions on the predicates, and that creates conceptual unity. And then the static unity, where all simultaneous relations must satisfy the static rules, and temporal unity, where all the states must be sequenced by causal rules. We'll see examples.

So let's start with an example here of a set of observations. What we are observing are two lights.

And the lights can either be at any discrete moment in time, either on or off. So here we have a sequence of observations. And one moment in time, the first light was off and the second light was on. Then the first light was on, the second light was off. Then both were on, etc. And I put the gray bars there to show that maybe the observations are incomplete. So at one stage, we can only see the second light.

Or there may be other lights or other objects that we do not see. But that's what we observe.

So you feed these observations in discrete time into the R-perception engine. And the R-perception engine searches for a causal theory that when applied to an initial condition, let's say that the

light A is off and light B is on, it will create a trace of recreated observations that cover,

is a superset, matches the initial observations. And if this happens, then our causal theory is a good one.

Now, the causal theory may infer the existence of hidden objects, hidden relations and whatnot. It may actually need to.

So here's an example of a causal theory that is generated by my own implementation of the R-perception engine,

because I re-implemented the R-perception engine as described in the paper by Richard Evans et al. And I ran it on this set of observations about lights on,

two lights, one on, one off at any point in time. And it came up with, it found a result. It found the result in 64 seconds.

It was a perfect match. And it actually invented a relationship, which it called Tread One, which we can, let's imagine that it actually means connects to.

And it found a static rule and a causal rule. It said that a light is on at any moment in time, a light is on, if a light that connects to it is off. And it found a causal rule that said a light turns off if it connects to another light that was also off.

So that's how the lights change over time, their status of on and off.

And it came up with initial conditions that said that, well, a connects to, first of all, that there's an object one, a light called object one that we don't see, but is there,

we imagine is there, that a connects to it, the light object one connects to B, and the light B connects to object one.

So that's the causal rule that it discovered.

Now, if we run this causal rule, produce a trace.

And as you can see, the trace matches the observation, it adds a new object.

So the coverage being excellent, being perfect in this case, and our causal theory is a good one.

It's actually a perfect one. It's not necessarily the only one, though.

So is this causal theory unified? So going back to Dr. Kant's requirement of synthetic unity of that perception, not every causal theory will do, though it may predict correctly, it may not be meaningful unless it is unified.

Dr. Kahn

Well, is it, we saw the four dimensions, the four dimensions of unification, is it spatially unified?

Well, all our objects are connected, directly or indirectly, to each other.

That's good.

So we have spatial unification.

Do we have conceptual unification?

So we have this new predicate.

We have two predicates, right?

Pred1, which we translate to connects to, and then the predicate that says whether the light is on or off.

Well, we have a constraint that says that a light can only be connected to one other light.

So pred1 has a constraint on it that says it's exclusive.

So an object cannot have pred1 to two objects, cannot connect to two objects.

That's a constraint that's a constraint that was discovered and part of the causal theory.

And also implicitly, the on relation, the predicate, has the value on or off, and it cannot be both at the same time.

So it's conceptually unified.

Is it statically unified?

Are the rules, static rules obeyed?

Well, for example, the static rule would say that given that B connects to A, if B is off, then A must be on.

So if you look at any place where B is off, A is going to be on.

And you could do that for every other light and relationships between lights.

So they all obey the static rule.

And the causal rule says, for example, here, that if B connects to A, then if A was off, then B must turn off.

So if you look at B, let's say here B was off.

Yeah, if, I'm sorry, if B connects to A, and yes, and if A was off, B must be off.

So if A was off, B becomes off, the next step.

So that's correct as well.

So statistically, we are true.

And temporarily, we are true.

We are unified.

And of course, that we get a thumbs up from Dr. Kant.

Our causal theory is unified.

Thus, it makes sense of the observations of the sensory inputs.

Now, it's no accident that Kant would figure in an active inference project.

There is a link between active inference and Kant.

And it runs through the celebrated 19th century German engineer, Hermann von Helmholtz.

He was a disciple of Kant.

He developed the theory of visual perception that operationalized Kant's epistemology.

And in fact, it anticipates predictive processing.

And in 1995, Peter Dayen and Jeff Hinton developed the Helmholtz machine.

The name is, and it is Hunter.

It is a type of artificial neural network that's trained to create the generative model from an original set of data.

And it can account for the hidden structure of the data.

So there's a link, which is discussed and elaborated in this paper, which is a very interesting paper.

All right.

So, close parentheses.

So we've looked at the perception engine from the perspective of Kant's philosophy.

So now let's look at it through the lens of machine learning.

We, the observations constitute a training set.

It's a very small one.

And the application engine is the learning algorithm.

And what is learned?

The output is a causal theory.

So the learning process is unsupervised logical inferencing.

And the output is a human readable logic program.

So we see here that there's some profound differences with the more popular form of machine learning in that the training set is really small.

That the product of the learning is actually a human readable artifact.

In this case, a logic program.

So this is the training set.

As inputted, lights, LED A turned off at time one, B turned on at time one, A turned on at time two, B turned off at time two, etc., etc.

So that's the training set.

And you feed this into the perception engine's algorithm.

And out comes a causal theory.

So in a little bit more detail, what the algorithm is and does.

First, it extracts the observed object, extent objects, the object types and predicates from the observations.

So we have on, we have object A, object B, we have LED as an object type.

So that's all part of the observations.

And that becomes part of the extent vocabulary.

Then the opposition engine imagines unobserved objects, types and predicates for the relationships and properties.

And that becomes a latent vocabulary.

So there's a step of imagination.

Then using the combined vocabulary, you know, both the extent and latent vocabulary combined, it looks for a unified causal theory.

So there's a set of constraints, rules and initial conditions that obey the rules of the constraints of synthetic unity of our perception.

Once it has this causal theory and with initial conditions, it applies the causal theory to these conditions and produces a trace.

It recreates observations, if you want, and augments them and extends them to the future.

Now it looks at this trace and compares it with initial observations for coverage and decides if this is a good causal theory or not.

Then it also looks at the causal theory complexity, how many rules, how complex are the rules, etc., and measures for complexity.

So if we have a choice between two causal theories of equivalent coverage, the app perception engine will select the least complex one using Occam's razor.

Now, if you look at this algorithm, you'll see that the boxes in green are not deterministic.

This is where search happens.

We can posit different kinds of objects.

We can find different kinds of rules.

So this is where search happens.

Now, app perception is implemented using logic inference.

Actually, it uses three forms of logic inference.

There's the one that we're more familiar with, which is deduction, where given rules and causes, we infer the effects.

Then there's induction, where given causes and effects, we look for the rules.

This is what science does, right?

We're looking for rules that would account for effects given the causes.

Then there's abduction, where given the rules and given what we observe, the effects, we're looking for the causes.

In this case, we're looking for the latent objects, the latent relationships between these objects.

And then you can combine both abduction and induction, where you're given effects, essentially observations, and you're looking for both causes and rules, which is what the app perception engine does.

And this is where, in the algorithm, these kinds of inferences are at play.

So positing latent objects, that's a form of abduction, imagining causes.

Finding the rules, well, that's clearly a form of induction.

And then applying the rules of a causal theory to some initial conditions to create a trace.

Well, that's deduction.

We have the causes, the initial conditions.

We have the rules, causal theory.

And then we produce a trace, the effects.

So that's deduction.

So the app perception engine uses all forms of logical inference.

And now just a reminder that the output of the app perception engine, that is what is learned, is actually human readable.

You may want to compare that to a large array of floating points produced by traditional, the more popular form of machine learning nowadays.

So here, this is what's actually produced by the app perception engine as it runs on a set of observations.

It produces a logic program that is human readable.

When you look at it, the only thing you need to kind of guess is what is meant by pred one.

And if you think, well, maybe it means connects to, maybe the lights are connected underneath a board out of sight of the observer.

But finding a unified causal theory is hard.

So we have to guess what the latent objects and predicates are.

What are the hidden lights?

What are the hidden relationships between lights?

And we have to discover what constraints might apply on the predicates.

And what are the initial conditions from which we want to recreate a trace?

What are the static rules that apply to simultaneous observations?

And then what are the causal rules that, given observations at time t , will predict observations at time $t + 1$?

This is hard.

As a matter of fact, it's non-polynomially hard.

The search space grows exponentially with the size of the input, which is the size of the extent and latent vocabularies.

So just like in chess, you can't predict to the end the consequence of a move because of common neural explosion.

With the app perception engine, you cannot systematically traverse the entire space of possible causal theories to find a good one because it's impossibly large.

So the job of the app perception engine is to find a causal theory in a ridiculously large haystack.

How to do this?

In my implementation, I follow the recommendations and I follow also the implementation in Richard Evans' paper by breaking the search space into chunks.

First, there's a region and the region says, so how many latent object types, objects, and predicates will we allow?

So what is the limit of imagination of the cognition actors that is trying to perceive a causal theory?

What are the limits of its imagination?

And within that region of bounded imagination, we carve it into templates where we say, okay, we're going to use these latent objects, types, these latent objects.

And so basically what vocabulary, specific vocabulary we're going to be using, we're going to use object one, we're going to use object pred one on top of the observed on predicate and observed A and B lights.

And we're going to use object one, and we're going to use object one, and we're going to use object to find a causal theory.

And we're going to use object to find a causal theory that fits this template.

So this is a carving up of the search space.

And having broken the search space into regions and templates, we have scopes in which to apply heuristics.

Now, why heuristics?

Because the systematic traversal cannot be done in reasonable time.

There's just too many candidate causal theories to look at to find a good one.

So we use heuristics.

We find ways of maybe getting to a good solution faster at the risk of missing it, but at least we'll have an answer or no answer in a reasonable amount of time.

And there's a number of heuristics that I have implemented in my implementation of the Perception Engine.

Well, there's time boxing.

At some point, you'd spend no more than this amount of time looking into a region or into a template.

There's multitasking.

Well, the problem is actually, as they say, embarrassingly parallel.

Well, you can explore multiple regions and multiple templates in parallel.

And so make good use of a multicore computer.

You want to make sure you don't repeat yourself.

So you don't want to traverse the same region twice or look at the same causal theory twice.

You want to satisfy.

Maybe a good enough theory is just fine.

We don't want to look for the perfect one necessarily.

We may not have time.

You want to fail early.

If you're in a region where nothing good is found, you may want to leave it quite quickly at the risk of maybe not finding a good one that is just over the horizon.

But you want to be impatient.

You want to throw the dice.

You want to go for the simple solution first.

You may not want to try everything.

Just sample some.

You want to start with the easiest part of the search space first.

Be judicious.

And so forth and so on.

And most importantly, be selective.

So reject any causal theory that would fail the constraints of unity of apperception.

With all these in place, running my implementation of the apperception engine gives pretty good results.

So here I did a run.

This is not cherry picked.

I decided to do a series of seven runs and collect the data and show it.

And in this run, I set up the apperception engine to only accept a perfect causal theory.

One that would produce a trace that totally covers the observation.

So I did seven runs.

The first one succeeded.

Found it in four seconds.

The second one, it took 102 seconds.

So there's some randomization in the order in which things are searched.

Luck is involved, as I said.

The third one, one second.

That was pretty cool.

The fourth one, well, took 204 seconds.

And then 96, 12, 99.

So quite a good distribution here.

Now I said, okay, I'm going to run the apperception engine again on the same training set that I showed earlier, those two lights.

But this time I said, I'm going to accept the theory that has at least 85% or more coverage.

So it predicts, it recreates the observations well enough, but not perfectly.

And I time boxed it to 30 seconds.

So you have 30 seconds to find it.

Go.

Go.

The first run, it find a causal theory with 75% accuracy immediately.

Then the same accuracy, same coverage.

10 seconds.

11 seconds, it hit 29 seconds.

It found a perfect one.

Then 11, it found 87% coverage and stopped right there.

That's good enough.

75.

Again, 100% is the first one it found above 85 in 18 seconds.

And 75% zero seconds.

So a good distribution again.

So we're getting into reasonable times.

We're not talking about hours here.

We're talking about seconds.

And I'm hoping to do further optimizations and bring it down to something even smaller so that a cognition actor can say,

I want to make sense of these observations, query the app perception engine, and get an answer, a causal theory, within maybe a couple of seconds.

That's my hope.

Now, something interesting here.

It so happens that what makes it hard for the app perception engine to find a good causal theory is formally equivalent to what makes cognitive science as a whole hard.

And that's this paper here makes the case and proves the case quite cogently.

So cognitive science wants to find models, functions, or algorithms that explain, account for situated behaviors.

So you feed into the cognitive science machine pairs of situations and behaviors, and you want to come out of it a model, an explanation, a function, or an algorithm that accounts for it.

Well, the paper makes the case that if the explanation is to be bounded in size, then the problem is computable, but it's not tractable in the sense, meaning that it's combinatorially explosive.

But once you have a solution, it is computable, intractable to verify that the solution is good, that it accounts for the data that you're trying to understand.

Well, this is equivalent, formally equivalent to what the app perception engine is doing.

My implementation was done in Prologue.

I will not go into the details.

It's about a thousand lines of Prologue.

I'll just say that Prologue is a programming language that uses deductive inference as its model of computation with backtracking.

So essentially, it searches for a solution and will backtrack if it took the wrong branch, if you want, and will look for a different way of satisfying a line of the program.

So let's just say that it makes traversing a search space.

We get traversing a search for free when we program in Prologue.

I won't go into any more details, but you can see some Prologue code here.

And it's the fun thing is that a program is akin to a logical description of the problems trying to solve.

I think it's very cool.

I think it's very cool.

And Prologue environment was augmented by something called constraint handling rules, which is an extension to Prologue that adds abductive reasoning.

So basically, in the program, you can say, assume this is true until proven otherwise.

And the rules, the CHR rules are there to verify if it can be proven otherwise.

So again, I'm not asking you to understand this code at all, but I want you to realize that this code is the code that actually executes causal theory, both the static and causal rules to build traces.

It is that small.

It's very powerful.

So combining Prologue and CHR, I found extraordinarily powerful.

I'm very excited about it.

I'm a programmer.

Next steps.

Well, next steps.

Now that we've solved individual learning by cognitive actors.

Well, I want to move to beliefs from sensations and to policies to validate or eliminate beliefs.

A lot of these beliefs actually fall out of our perception.

Latent objects and latent relationships and properties can be considered as beliefs.

Then there are other kinds of beliefs that can be obtained from what's been perceived.

There will be introspective beliefs that communicate how the cognitive cognition actor is doing in terms of competence, prediction area rates,

how well its perception is doing, and whether it is engaging with other cognition actors.

Is it relevant?

I will have feelings, which will provide normativity to these beliefs.

So if feelings are signals of risk to homeostasis, loss of resources, physical damage, too many prediction errors.

So that's anger, pain, and fear.

And feelings will taint beliefs over time.

And tainted beliefs, good beliefs, bad beliefs will want to be eliminated or validated through policies that will be synthesized by the cognition actor.

And each cognition actor will emit, make available to others its API, what predictions can be made about the beliefs of this cognition actor,

what actions are available to others to be asked of the cognition actor.

And then as cognition actors connect to one another, as the cognition actors form the umbel of other cognition actors,

then the cognition actor will be able to predict the beliefs of others,

will be able to compose policies made out of actions that are implemented by other cognition actors,

and eventually we'll have a society of mind, which is a bunch of intersecting boom belts.

So that's it.

So I see what the society of mind is a complex system of collective theorizers.

And I'm going to try going further with this project to answer the question,

if a collective of theorizers can self-organize to actively sustain itself.

So thank you to the Active Inference Institute for inviting me to present and for providing a home for this

project

and for the constant support and encouragement.

I'll see you later on Discord.

Thank you.

Awesome.

Thank you, JF.

Just to conclude the session, I'll read two questions and let's maybe address them in an upcoming Robotics and Embodied meeting.

So if you're excited about this project, certainly we all are, and about symbolic active inference, join the Discord and participate in the Robotics and Embodied.

But I'll drop these two questions from David Williams in the chat who wrote, one, how important is conducting this work in real world versus simulation?

And two, what tools or components are missing in the Robotics Toolkit to make this research easier and better?

I know those are things that you have a lot of thoughts on, so I'll look forward to discussing with you more.

Thank you, JF.

Thank you.

Peace.

All right.

See you.

And it's a great segue from collective behavior in surprise minimizing agents to collective behavior in surprise minimizing agents, I guess.

Welcome, Connor.

Hey, sorry.

I just turned on my audio, so I just heard the last thing you said.

Yeah.

Hey.

Welcome.

Good to hear you.

Yes.

Thanks.

We're looking forward to your presentation, multi-agent, active inference and multi-scale alignment, current developments and challenges.

So feel free to share your screen or proceed however you prefer.

Great.

Thanks.

And before I start, can I just, because I've had issues with my voice, like saturating, like my microphone saturating.

How's my audio?

Is it clipping at all or does it sound okay?

This is good and I'm watching it.

Okay.

Perfect.

Thank you.

I will share my screen.

Let me share this one.

And then I will...

The one thing I'll note.

Oh, yes.

Okay.

It's just JF symbolic implementation does not use statistical distributions.

It uses the symbolic and the logical inference.

And now we're going to move back into the distributional space and it will be awesome to see similarities and differences.

So thank you, Connor, to you for the presentation.

And thank you, Daniel, for the introduction and for inviting me as well as big thank you to the other organizers of the third applied active difference symposium.

Yeah, I'm really happy to be here to present.

So my name is Connor Hines.

I'm a PhD student at the Max Planck Institute of Animal Behavior.

And I'm also a researcher at the versus AI research lab, R&D lab.

So I'm going to do something a little bit, I guess, unconventional for people in my position.

Like I'm a junior researcher coming to the end of my PhD.

So usually when I give a talk, I will present on my own research, like what I've been up to for the last 10 years working on.

But instead of that, I'm actually going to talk about, given the motivation of the symposium, I'm going to talk about something that's more of an overview or perspective on the current state of the field in multi-scale active inference or multi-agent active inference.

And what, in my opinion, we need to do to move forward as a field.

I think that's very resonant with the kind of motivations and the title, indeed, of this symposium.

So I'm going to give a general analysis of what multi-scale active inference is, why it's important.

I'm going to provide a brief analysis of its formal basis as it currently stands.

And then what we need to develop in this kind of sub-discipline of active inference, multi-scale active inference, to really make it rigorous and really to actually reap the benefits of what it promises.

So generally, active inference has been used a lot to design agents that can solve problems, plan, and just generally emulate behavior that we deem intelligent, which includes things like risk-sensitive decision-making, intrinsic motivations to resolve uncertainty.

And finally, from a more scientific standpoint, the ability to furnish a process theory about how biological brains actually might work.

But in a lot of the theoretical work on active inference from the last 10 years, or 10 plus years, really, there's also, alongside all the kind of practical building adaptive agents, there's a claim that active inference is inherently or intrinsically multi-scale from the very get-go.

It is a multi-scale framework.

It's not just about building single agents.

So it's really, whenever we write down a single active inference agent, what we're implicitly implying is also a nested hierarchy of active inference agents, both below and above.

So colloquially, you'll often see this in papers as the idea that there's Markov blankets all the way down.

Markov blanket is a statistical structure that's very intrinsic to the definition of agents as they are defined under active inference.

So I'm not going to get into defining that.

I'm kind of assuming that there's a more disciplinary audience there, but I'm sure other talks from symposium can provide better clarity.

So yeah, Markov blankets, active inference all the way down, all the way up.

And at any given scale, cruciate, the free energy minimizing dynamics or the active inference dynamics are kind of claimed to be aligned with or parallel to the free energy minimizing gradients at the level below and above.

Markov blanket is a very important thing.

So what the claim is that as agents are doing their thing and doing active inference at one level, it both entails and is constrained by active inference processes of the macro agent that they're participating in.

Markov blanket is a cell that's part of a tissue as well as the micro agents that comprise them.

I am a free energy minimizing cellular agent comprised of organelles that are also minimizing free energy.

Markov blanket is a very important thing, but it's a very important thing to do with the micro agents that are being able to do with the micro agents.

So this kind of constrained, neat nested gradient descent on free energy is part of the story of multiscale active inference.

And it also crucially assumes that these dynamics are aligned, correlated, cooperative across these different scales.

So I should mention that there is a formal argument made more recently, I would say in the last five years about how this is possible.

And it relies on an apparatus from statistical physics called the renormalization group.

This basically allows you to analytically identify shared symmetries energy and conservation laws at different scales in a given system that's comprised of subsystems and subsystems, so on, at infinitum.

So there's a formal argument specifically made in the free energy principle for a particular physics monograph from 2019 that applies the renormalization group apparatus to multivariate stochastic differential equations that are kind of the equivalent of agents.

So you can apply that framework to certain sorts of coupled stochastic differential equations that exhibit Markov blanketed squash coupling structure.

And you can kind of prove analytically that there are going to be nested systems of Markov blankets, and that they're all in some sense minimizing free energies at their own scales.

So I'll get more into that argument later.

But I just want to mention that as I define multi scale active inference that that there is a form argument that that's related to it.

So this slide I just put together to demonstrate the idea of nested free energy minimizing processes visually.

So at a given scale, we can think of an agent as occupying some point in its free energy landscape indicated by this red orb, which represents say its configuration, its beliefs and its actions.

And it performs active inference and in doing so minimizes its free energy.

So it changes the position of that ball on that landscape.

And that is all we need when we say active inference that that corresponds to the agent doing inference and doing action and kind of getting to the fixed point of its local free energy landscape.

The multi agent case is simply when we add more of these processes.

So there's other agents usually assumed to be similar agents and the word similarity, let's put a asterisk on that.

And they're all sitting at different points in their own free energy landscapes.

They're all the position of their local red ball is maybe in a different place.

So the claim of multi scale active inference is that as we link these multiple active inference agents together, so they can actually exchange information like observations and actions with each other.

So what we automatically get is some kind of super agent that is also minimizing variation of free energy and is in some sense an emergent or supervenient active inference agent.

And I say the word we automatically get a super agent with an asterisk because there may be some conditions on that mapping from local to global that have to be elucidated.

So we'll come back to that in a bit.

But in short, I think the definition of multi scale active inference is very eloquently put in this paper by Raphael Kaufman, Pranav Gupta and Jacob Taylor.

I think Raphael Kaufman is actually going to be on the panel later.

And this line from their paper is a really nice, I think, just summary of what it is.

So I'll just read it out loud.

The upshot is that in theory, any active inference agent at one spatiotemporal scale could be simultaneously composed of nested active inference agents at the scale below and constituent of a larger active inference agent at the scale above it.

In effect, active inference allows you to pick a composite system or agent A that you want to understand.

And it will be generally true that both that that agent A is an approximate global minimizer of free energy at the scale at which that agent reliably persists.

And that agent A is composed of subsystems, A sub I, that are approximate local minimizers of free energy.

So that is the claim as I'm going to continue evaluating in this talk.

And I think it's just a great reference point to make.

So, okay, that's what multi-scale active inference is.

Why is it important?

Why do we actually care about that?

That sounds philosophically nice and beautiful visually, but why is that important?

So there's a ton of actually really important implications of this, both for the engineering and the natural sciences.

First of all, the namesake of the symposium, I assume inspired by this recent paper by Carl Fristin et al. about enacting ecosystems of, you know, parenthetically shared intelligence.

So this is the third applied active inference symposium.

So to really make this resonate with the applied aspect, let's make this very concrete.

If we can figure out this multi-scale endeavor, then we can actually engineer distributed systems of multi-agent intelligence where local agents

and doing their own little local active inference processes are also cooperatively instantiating a global agent that's also performing active inference.

So this has huge computational potential, of course, compared to kind of the state of the predominant methods for artificial intelligence, which are deep learning, which really is about propagating global information through an entire computation.

So although you could argue back propagation is local in some sense, it's really not local in the way that multi-scale active inference promises to be local.

So if we can figure out how to actually engineer multi-scale active inference, it will have really tremendous implications for the study of artificial intelligence, just from that pure engineering standpoint.

It'll be cheaper in one way. It'll be computationally, energetically, memory-wise, cheaper, a lot cheaper.

Secondly, from a kind of more natural sciences motivation, which is kind of where I'm coming from, I'm doing a PhD in biology.

So I'm interested in questions about actual real systems in nature.

Just the idea of being able to get super specific and rigorous about phrases like emergent intelligence, emergent computation, collective intelligence, super organism, that's often thrown around.

We're talking about social insects, right? These are terms that you hear thrown around in many different scientific disciplines that deal with multi-agent systems, network systems.

But none of these terms, to my knowledge, have really rigorous or precise conditions.

Multi-scale active inference is a kind of framework that's in the position to provide those rigorous definitions and conditions.

So from a scientific standpoint, it could really lend a lot of potential and usefulness for other scientific disciplines.

And finally, another pragmatic motivation. There's loads of fields that are obsessed with designing and engineering systems where local selfish individual behavior can lead,

when networked appropriately, to some desired collective outcome. And these disciplines, you really want to figure out how to engineer that properly.

So this goes from the design of financial markets and trading systems, all the way down to how do you design a multiplayer video game.

So that's kind of just motivating why is multi-scale active inference even interesting.

So then the question, of course, becomes, is the multi-scale active inference claim actually true?

Are all multi-agent active inference systems comprised of and themselves comprised nested hierarchy of free energy minimization agents?

A glance at a smattering of other scientific disciplines that specifically deal with multiple agents, multiple interests, collective phenomena, like coordination, group behavior, collective intelligence.

A glance at all those disciplines would naively suggest that the answer is no.

So there's things like frustration in thermodynamic systems, game theory, the very existence of zero sum games and Nash equilibria, bandwagon effects when we're talking about social networks and opinion dynamics.

So there's things like sacrifices for the common good, which we see in different contexts in biology, like in the context of kin selection, but also in the context of arguably cell cell death in a tissue.

So these are all basically plenty of systems where local constraints and global constraints or desires or free energy gradients, whatever you want to call them, come into direct conflict.

So the obvious example that I listed at the top of these bullets is the idea of geometric frustration that we see in Ising systems with very low temperature.

So these Ising models basically describe lattices of ferromagnets that are happy when they're pointing in the same direction as their neighboring ferromagnets, and they can be in an up or down state.

So basically the magnet can be pointing up or pointing down.

And these global systems are defined by a global energy function, and the whole system is in some sense trying to minimize that global energy function.

But sometimes you'll find cases in these collective systems where this little spin in the middle cannot be happy because they're getting conflicting information from two neighbors.

I want to be pointing up in blue like the agent on the left, but I also want to be pointing down like the agent on the right.

So this is a system that's collectively finding some fixed point of its global free energy, but it's actually leading to a local conflict where this agent is not at a point where it can do anything to make itself happy or minimize its free energy further.

So just from even the basic glance at ferromagnetic lattices, we already see instances where local and global gradients or local and global optima are not aligned in the right way.

So given all this, the burden of proof for multi scale active inference is still on us.

So we need to show that collective active inference systems generically do align again across scales.

And maybe if we can put an X across the word generically and it's not some automatic condition, if they don't, then at least we have to establish exactly the conditions in which they do.

So anecdotally, we do actually have some conditions.

There are some, seem to be some kind of basic ingredients to get collective active inference to work.

So one is that we basically need agents to exchange actions and sensations across some kind of Markov blanket.

This is not really a condition.

This is almost more part of what it means to be an agent.

So having Markov blanket separation between agents is just another way of saying we have multiple agents in our system rather than a single agent.

If you're violating the Markov blanket property, so internal states of one agent are not allowed to see the internal states or external states of another agent, then you're kind of cheating because you're kind of saying it's actually one agent.

And what you're doing is information sharing within the brain of a single agent.

The second condition, which is something that's often elucidated more anecdotally and not really formally, is this idea that agents need to have some kind of shared narrative or shared hidden states or sensor space in their generative model.

So I've worked a lot on collective active inference systems, just simulating agents and trying to get them to do interesting things together.

And my intuitions and experience do agree with this basic fact.

If the agents don't have any similarity in what they're representing or trying to achieve, then it's kind of like trying to fit a square peg into a circular ball.

So this is really nicely elucidated in the one of the earliest cases in this paper, A Duet for One by Fristin and Frith from 2015, where they show that for two agents to really align, they kind of have to have a shared generative model.

And then you can get kind of this nice synchronized behavior.

Again, these things like what does similarity mean?

What is a shared narrative actually made formally mathematically?

Those things have not been elucidated yet.

So right now, a lot of the building of these collective systems is based on our intuitions and kind of engineering things in using some vague guidelines like, oh, they should have a shared sensor space.

But there's no mathematical conditions or guarantees about what degree of similarity is needed between two agents models to get the intended dynamics.

And finally, we have to have at least some agreement between the generative model of each agent and the generative process, which is really the behavior of the other agents generating their data.

So this is kind of related to the previous point about having shared generative models.

But just to be very specific, the physics of the space that transfers your actions to my observations, that physics can't be dramatically, crazily different than how our generative models represent those physics.

So if we took two fish with the same generative model of each other, and they normally would school together in a fish tank, but we throw them in a volcano or shoot them out into outer space, they won't school together because then the generative process is so dramatically deviating from the way they are representing that physics, the way their generative model is constructed.

So these are, again, just ingredients, kind of guidelines or anecdotes, but there's nothing really rigorous behind these conditions.

They're more like a list of best practices.

So now let's get on to actual rigorous stuff.

So the first real rigorous attempt to show that multiscale active inference generally works is in one section of this free energy principle for a particular physics monograph from 2019.

So let's use this apparatus I mentioned earlier, the renormalization group operator, to basically show how one can successively coarse-grained multivariate stochastic differential equations that admit sparse coupling between their state variables.

So the main result in my mind that connects these renormalization results to multiscale active inference is the fact that the Lagrangian of the system at one scale can be expressed as a function of the Lagrangian at other scales.

And that applies in a scale invariant fashion.

That is the main, like, kind of output or the main benefit of using a renormalization group apparatus.

So you can kind of think of Lagrangian like the generative model.

It's a physics term, but it's related to the generative model of the agents that comprise the system and therefore also their free energy.

So in terms of active inference, it means that this reasoning of the renormalization group can be used to smoothly move between the models of individual agents at one scale and the model of a collective or larger agent at a different scale or a smaller scale for that matter.

And the nice thing about it is general for all kinds of dynamics and that's generally models.

It doesn't depend heavily on the form of the stochastic differential equations that form your system.

The issues with it is that there's not a global link to Bayesian mechanics and active inference.

It's still all done in the traditional physics form formalism.

So we don't actually have an explicit link to local inference and global inference.

Although if you know the connection between the Lagrangian and the generative model and the free energy, then you can make that connection.

But it's not actually made explicitly for us in this part of the monograph.

It also requires the assumption that the generative model and the generative process are identical at the local level.

That's related to how the Lagrangian is defined.

That's also a restrictive assumption. That's probably not realistic in my opinion.

And then finally, there's something about kind of spatial temporal segregation of scales.

So we need to make assumptions about how fast random fluctuations are at one scale relative to another scale in order to justify kind of course training or forgetting about certain states as you move between scales.

And that's also something that you could argue current research into collective dynamics challenges that assumption about how fast noise is at one scale relative to the next.

So now I'm going to discuss quickly another small contingent of active inference research that is attempted to address this mapping between local and global inference processes.

So what I want to kind of just generally say with this presentation and to our community is that the kind of approach taken in these two papers.

First of all, active inference model of collective intelligence and spin glass systems as collective active inference.

This is one of the types of research I think we really need to move active inference multi scale active inference forward.

So I'm not trying to be too biased because I am the first author on the second of those papers, but I'll also be the first one to point out the limitation.

But benefits wise, I think these approaches are really important because they formally relate a local generative model at one scale to a global generative model at a different scale.

So really tie how did the parameters of one model relate to a coarse grained model.

And these are really good steps in the direction of a formal theory of collective intelligence that goes from local intelligences to global intelligence.

However, there's still issues with these.

One of them is that they only deal with issue with inference at the global level, not active inference.

So these both these papers concerned with a bunch of local active inference agents that cooperate to form a global inference agent, like a passive Bayesian agent, rather than an active inference agent.

And it's also unclear whether the system studying these papers are actually very generic, like the results are generic to study collective intelligence in general, or they're nice formal arguments, but they're only applicable to these specific systems.

So we still don't have something that's even more kind of zoomed out and abstract than these, which tend to be a little bit case specific.

And finally, the actual scale transcendence that we're doing in these papers is still relegated to really one step.

We're not doing the full multi scale infinite scale regression.

That's something like re-normalization with promises.

So that's kind of the current overview of what I think are the most promising directions in multi scale active inference.

And I'm aware on time, sorry, five minutes.

So I'm going to quickly try to go through what I think are really promising directions to push in terms of multi scale active inference and kind of intuition parts that I think will help us study these systems in a way that's different and also actually better informed by other research disciplines.

So the general idea that I'd like to put forward is that misaligned gradients can actually be a good thing.

So it's actually sometimes good when local free energy gradients are misaligned with global gradients.

So sometimes the global system will actually do better if the local systems are performing worse.

So this is something you could call multi scale conflict with a free energy minimizing process at one scale are actually doing bad, quote unquote, but it's because they're being driven by some higher scale process that is doing well.

So rather than trying to always avoid constructing processes like this, I think this kind of frustration to use the analogy from statistical physics can actually be an inspiration for what we should investigate further because it actually might be key to facilitating optimality at different scales.

So the reason I put this forward is because there's loads of research just recently researching the last several years that are suggesting that actually making local agents more frustrated or more unhappy might coincidentally or not coincidentally lead to better collective or global outcomes.

So this is expressing various forms in various bodies of work.

One of the biggest patterns I've noticed is the study of collective behavioral systems over the last several decades.

The idea that local noise and local dysregulation can often facilitate global coherence or global coordination and where multi scale active inference has something to say in my opinion is in framing this benefit of local frustration in terms of a misalignment of free energy gradients.

So maybe that actually temporary misalignment of local free energy gradients from global ones may facilitate the descent to fixed points in the global free energy landscape that satisfy everyone at all scales.

So basically expressing idea that's been known in various communities like stochastic optimization and stochastic resonance theory for decades.

But I think we as active inference practitioners have a new and potentially useful perspective to shed on that using the language of active inference and free energy minimization and basic inference.

In general.

So instead of thinking of accelerating optimization by just adding noise to the system, we can think of exactly how to design local generative models such that there's an optimal misalignment of local and global gradients or local and global generative models in a way that facilitates everyone in the long run actually facilitating or minimizing your free energy.

So yeah, that's just kind of like something I'm putting out there.

But I just want to put that out there in this venue because I think it maybe will inspire other people to think in a similar way.

So just to conclude now, multi scale active inference, I would say is still largely based in theoretical or philosophical descriptions and illustrious simulations, but we're still lacking a formal theory.

There's some theory in terms of the renormalization group arguments of the monograph, but they're still, in my opinion, a bit underdeveloped, a little under demonstrated and rely on some restrictive, restrictive assumptions like the fast and slow fluctuations, the identity between generative model, generative process.

There's a few more recent papers, those two by Kaufman et al.

And then by myself on Spengler systems that have attempted particular proofs of multi scale based in infant systems in particular situations, but their generality is still not known and not proven.

So we, what I'm kind of trying to conclude with is by saying we need to incorporate findings from other disciplines related to the role of noise, conflict and frustration in facilitating, not subverting collective intelligence or collective coordination.

And I think we can really benefit by looking at these other disciplines to help us build a really powerful formal theory of multi scale active inference.

And finally, I think we need to set the goalposts for what counts as a formal proof of multi scale active inference.

And once we get there, once we say, okay, this counts as proof, this is satisfying, how can we use that to actually do the hardest part in my opinion, which is engineering actual multi scale active inference systems that are intelligent and minimizing free energy at multiple scales.

Um, yeah, so with that, I'm going to conclude, it looks like I'm just on time.

So yeah, thank you again for the invitation to present and I'd like to thank a bunch of people who are listed here and beyond who have influenced my thinking and kind of my opinions.

Um, yeah, if there's time and happy to take any questions, of course.

Awesome.

Great.

I'll just give a few seconds if anybody wants to type in a question.

Also, it's really cool.

Like Aswin in the previous session was highlighting PI MDP and just the way in which we enact the collective intelligence, different people seeing a paper where an analytical formalization is introduced, and then there's still so much work to get it to the package and then so much more work to take it to the last mile.

And I think your presentation really checked a lot of those boxes.

Um, I'll just read a question.

And then, um, that will just be an appetizer for our continued discussion.

So Marco Lin asked, do you expect the inferentially connected dynamics to exhibit behavior akin to theories of multi-body systems?

And to what extent can we transfer insights from that multi-body of work?

And then second question just for our thinking and learning from Marco.

Have you explored integrating work with, uh, integrating work on self-organized criticality with multi-scale active inference or other frameworks who can provide more flexible frameworks or assumptions for a generic notion of multi-scale dynamics?

Great questions.

I hope that we can continue, um, have you back anytime or just continue to collaborate in the ecosystem.

So thank you for the epic talk, Connor, and good luck finishing your PhD.

Thanks a lot, Daniel.

Yeah, I'll talk to you soon.

Talk to you soon.

Talk to you soon.

So thank you for that.

All right.

Greetings.

All right.

Well, our next session.

Hey.

Hey, hey Bert.

Greetings.

Great.

Hey, how you doing?

Good.

Good.

Good.

I'm good.

I'm good.

I'm good.

I'm good.

I'm good.

I'm good.

I'm good.

I'm good.

I'm good.

very well um our next session is with bert devries dimitri bagaev and bart van erp it's going to be called towards user-friendly design of synthetic active inference agents and i know a lot of people are super excited to see this really practical and cutting-edge work so to you bert and just let us know how we can support okay great uh well is my audio good yep sounds good okay then i'm going to share my screen uh i hope i picked the right one i don't work it's zoom quite often looks good it looks good yeah all right super um well thanks a lot uh daniel for for hosting this uh symposium i've been watching some talks it's really amazing and uh we really we feel privileged to get the chance to present ourselves so we are also just like a few others before us interested in developing a toolbox for active infants and so this picture or it kind of shows what we're um what we're about or what we're interested in so here's a lady on the left hand side and um i'm gonna try to get a laser pointer and she has this idea about a rewarding behavior for a vacuum cleaning robot right so she's writing down she has a textual expression move around the apartment apply suction until the floor is clean do not touch objects and when done return to the dog so that that's not so hard i'm gonna rate that with one star out of three stars in terms of difficult difficulty level to specify that um but that's not enough to program this robot right because what she really needs to do now is to specify a generative model and uh that you know the there's effectors and actuators right the robot has to move around apply suction until the floor is clean so the sensors probably a camera uh do not touch objects so maybe there has to be object recognition this is a really difficult task to come up with this generative model here and on top of that she has to specify this kind of rewarding behavior in terms now of probability multi-distributions of this generative model so very difficult i'm gonna rate that with two stars because the next thing she has to do for this model is to specify the inference procedure to do actually active inference and free energy minimization in real time for this complex model and really that's almost impossible right only a few specialists can really write a procedure for for variational free energy minimization in some very difficult model so what we are about what we've been working on is to try to automate the inference task so get rid of the three stars and yes she will still have to specify a model but in the long term we try to get away from that so in the long term we hope we will get a toolbox and now we're talking five ten years right um where a textual description would be enough to specify some initial model with an initial prior and everything else is just automated inference learning of states parameters structural adaptation of the model even maybe based on her feedback updating uh the prior so that's long term for now we would be very happy if we could just automate the inference uh task so um why why is it so difficult to specify inference for an active inference agents well we have so many competing uh kpis right

we we uh we want to do this for large model scopes not just for a b c d models but maybe there's also continuous variables and our logical models right it must be very user friendly we really don't want her to worry about robustness um of her code um we don't want her to worry about whether um two variables are have conjugate relationships right um adaptivity we want to update states parameters maybe even the model the model structure has to be low power because these engines often run on edge devices all right so they run on their battery powered that's to be in real time because you can't learn how to ride a bike if there's no real-time reasoning and on top of that you actually want to minimize variational free energy right you want to do it at least as good or at least in the neighborhood of if you would do a manual derivation and some of these this is a rat that bite each other right if you uh you want to minimize variational free energy but you have to do it in real time and on low power yeah that kind of bites each other right so these are difficult kpis that we we want to yeah we we they're all important you can't just take one out um because then the whole system wouldn't work um so um when you read papers on active inference you often also read and now we implement variation of range minimization and that can be done by message passing on a graph and i want to clarify first why it has to be done by message passing on the graph um i'll do that by giving a very short answer and then do an example the short answer is that bayesian inference involves um computing very large sum of products like what you see here on the left hand side here's a product $ac ad bc$ and then we sum them in ac plus ad and so forth this is a sum of products now we know by the distributive law that this here on the left hand side can also be computed as on the right hand side if i multiply this out i get a times c plus a times d and so forth this is a product of sums and they are exactly the same thing the only difference is that to compute the left hand side takes four additions sorry four multiplications and three additions to compute the right hand side takes two additions and one only one multiplication so on the right hand side is much cheaper to compute than the left hand side normally when we write down marginalization and bayesian inference we write things down in the letters in the left hand side what message passing does on the

graph it will automatically convert that into much cheaper to evaluate product of sums and i'll give an example of that um so here is an uh an example model f of seven variables $x_1 x_2$ through x_7 and this model happens to be factorized f_a of x_1 f_b x_2 and so forth now we can draw this factorization in as a graph and what we do and this is called a forney style factor graph what we do is for each factor f_a we allocate a node so f_b gets a node and f_c gets a node and we associate the variables in our system with the edge and an edge is connected to a node if that variable is an argument of that function so f_c is a function of $x_1 x_2 x_3$ and that means that f_c connects to the edges $x_1 x_2 x_3$ and f_d is only function of x_4 so f_d only connects to the edge x_4 so what you can see in this graph is this graph is nothing but a visualization of the factorization assumptions that we have for this model now if i'm interested in a big marginalization uh task uh integrate out over all variables but x_3 so $x_1 x_2 x_4$ and so forth through x_7 i'm interested in this um then making taking advantage of this factorization i can rewrite this sum this basically this um sum of product into um a product of sums as as below here what you will see here below this computes exactly the same thing but i've made use of

[illegible]

or the generative model then q of b the variational posterior for this node will also depend on x_1 x_2 x_4 and on no other factors if you just do that then you will count some of the variables double because x_1 is part of the belief over f_a but also part of the belief over f_b so we just have to discount that by dividing by beliefs over edges that means that I make now an assumption that my posterior beliefs is is is divided into local belief is over nodes and local beliefs over edges over variables this will make things a lot simpler in fact if my graph is a tree and it is a tree here and I would do message passing on that tree and I could suppose I could do that perfectly everything is linear gaussian then I get perfect bayesian inference there is no approximation so this is a good assumption sometimes it's still very hard to compute a message because even the the the single messages that come out of these nodes they're still integrals or summations and in particular the integrals may be a problem we may not have an analytical answer so what we sometimes do is add additional um assumptions we'll say well the posterior belief over f_d I can't compute it in general but I'm going to just assume now that it's a gaussian that makes it easier or we can make an extra factorization assumption and say the posterior belief over f_b which is really a belief over the joint x_1 x_2 x_4 is going to be broken into uh independent belief over x_1 and in belief over um x_2 and x_4 um these additional assumptions if I impose them as well this is what I recall now if I if I if I if I all substitute that here in q of x I get what's called a constrained bethe free energy this is the same death as in the oppenheimer movie this is hans bethe where it's uh named after and um so we have a graph now

that um that is highly factorized um and we have um local beliefs over nodes and over um over edges and they're indicated with with red and we have additional constraints in green they could be gaussians or mean field constraints or other constraints and now we will assume constraints that make it possible to compute all the messages and now we can just automate this by making different assumptions we can turn this into expectation maximization or belief propagation or um hybrid forms draw we can turn it into any relevant message passing algorithm that you've heard of so this is a very nice umbrella framework that uh basically encompasses everything and um and there is a we've written a pretty large paper paper on this um in the entropy journal where you can read all the math on how this works so we've talked about why message passing namely because it's the most effective way of doing it inference and we've talked about which messages to compute namely we turn our variational free energy into something called a constraint better free energy and then we can compute messages the only thing that's left is well when do we pass these messages right what is the sequence of messages which one comes first and um this is where we see a lot of papers right um you have to write control flow what's called control flow you have to say okay here is my algorithm for active inference first I specify a model then let's do inference for every time step collect a new observation update the state update the desired future right and so forth compute expected free energy select the policy etc um this kind of program the problem with active inferences is that there is nested for loops in here here's a for loop and here's another for loop and for each of these policies I'm going to have to go into the future so I'm going to have another time loop so it is four loops in four loops in four loops this will completely

explode in terms of computational complexity so as a result some very clever people have written very clever algorithms of doing this much faster sophisticated inference uh branching time active inference dynamic programming

these are recent proposals for doing this very clever in the end all of these proposals come down to a particular just a message passing schedule um once we commit to message passing on the graph as our inference procedure it's the only thing that's going on

and all of this sophisticated inference and branching time active inference all it does is it schedules the messages it says first this message then this message then this message

i don't mean that as a slide to this message i don't mean that as a slide to these uh algorithms they're very clever and as we've seen in the presentation by as

been paul you get huge improvements if you go from regular inference to sophisticated inference but it's good to realize that

these algorithms just specify in the graph which message comes after which message um so here's an example

uh here's an example of a graph and a message um sequence here's a message one then message it's two and message three goes up and then we go from fc to f f and here's message five and then we go to fe and this could correspond this sequence to

are the dynamic time programming efe or sophisticated inference um there are a couple of problems with this approach which we basically with with having the user to specify a clever algorithm first of all you have to be a specialist to do it right only these are very clever people so so that means that um if we let it leave it to um let's say to an engineer and a company it's it's well it's the high probability is not going to get it right that's very unfortunate um but there is another issue and that is that in it in a sense it's a global variable um in the message passing schedule all nodes are visited because if a node would not be visited then we shouldn't have it in the graph and that means if one node crashes basically the message passing schedule is invalid i have to reset my system and if you fly a drone if it's deployed and it's out in the field and a node crashes a transistor burns out and i have to totally reset now my system i have to compute a new message passing schedule um then you're not doing inference and you know and your drone flies into the wall so this is not robust and it also for very this the same reason we may actually want to uh take out a node we might we may want to prune a node we want to want to do a structural adaptation and we can't do structural adaptation because we have to reset the system recompute a message passing schedule

so this procedural style where an engineer a priori specifies which message comes after this message has some disadvantages it's not very robust and it's if you want to do it very clever you have to be really really a specialist so a better system is what we call uh reactive message passing and um it's very related to what was in the first session uh called the the actor model uh keith dugger had a nice presentation on the actor model so what we will do is we will say we will not have a global message passing schedule the engineer will not specify anything anymore the inference code that an engineer will have to write is just say react to any free energy minimization opportunity in other words there is no

inference code it's completely automated and we will replace this global message passing schedule by local triggering inside the node so each node is now just an autonomous autonomous system that's interested in minimizing its free energy uh it can do so by sending out messages and when will it do so well it it receives messages and when it feels when it looks at these messages and it feels like oh there is an opportunity for me to minimize free energy by or expected free energy by sending out a message then it will send out a message

and each node will do so by itself asynchronously so you get parallel distributed processing or concurrent processing as keith called it um in principle you could play this game on many uh computers at the same time uh and so you get tremendous uh advantages um first of all you don't have to write difficult code second of all you can do multi-threading um or you can you can run it on multiple computers at the same time and so and there's also um robustness um um uh advantages because if a node crashes then there's nothing that stops the system from just finding another path right if if this node crashes this path from here's message three this path now doesn't work so i cannot send anything to fe anymore from uh no from x well then i just sent a new message here why not it's like when water falls down a mountain and it zigzags its way down into the valley and you have where you put up an obstruction it's just finds another path um not the preferred path this has to find well the the second the second best path because the first path has been obstructed right and that's what's going to happen in this system as well right that's just how nature works it tries to find the best path the easiest path and if that's not available then we do the second best path and that's also what you can do with reactive message passing so you can prove notes you can do structural adaptation um and it's far more robust and you can also get uh let's do um chance encounters with other drones right drones that get close can start communicating with each other and when they're far away they stop communicating with each other and this is no problem because you can basically change who chains um notes can change on the fly who they communicate to and who they want to listen to and uh so that's that's the way data works and also how it works when we do uh reactive um programming and react message passing so in summary um we're interested

in automating inference in um active influence agents right um because it's an operation that's basically only for for experts and this active influence technology is not going to be successful unless we get more people uh let's democratize it and we get competent engineers being able to develop good agents right you shouldn't have to be a top specialist in the world to go to develop an active influence agent now in order to automate inference you must do message passing and i've talked about that for efficiency i've also talked about which messages to pass not necessarily do you have to follow this framework but constraint better free energy framework is very convenient is an umbrella framework that basically um goes over all the interesting other uh message passing uh computations when s is passing reactive message passing it's fully automated so you don't have to write any code anymore in principle it's it's you can do parallel distributed processing it's robust to structural changes you can learn new inference pathways so lots of lots of advantages here now how do we do it um i like to introduce a toolbox that you've been working on uh called rx infer and we do that with my lab here at the university i'm here in eindhoven in the

south of the netherlands and we have a lab the labs called bias lab here are postdocs and assistant professors and phd students and we've been working on this for many years and some of these uh like albert and ismail and thais have written dissertations um and our best work we have consolidated that in a toolbox and the toolbox is called rx infer and you can if you want to have a look you can go to the website arctonfer.ml and i'm sure though rx infer works uh in the way that i've just discussed it does message passing it tries to minimize constraint better free energy that means it it can come up with all kinds of uh message passing algorithms um um it will do it in a reactive way and uh it will try to do it in real time and low power and all the kpis that we're talking about now it's of course not done but it's functional and i'd like to show some demos and i will uh leave it to dimitri and bart who are two advanced phd students in my lab to show the demos so i'm gonna stop sharing

awesome thank you bert great talk sure uh can you hear me uh yeah yeah okay i will try to share my screen okay so you should see it now looks good okay so yeah hello uh to everyone uh i'm uh dmitry bagayev so i'm a phd student in vice lab in aynhavon university of technology uh and yes i have a small presentation about uh actual software developments in bias lab and yeah so over the past few years we have significantly improved our tools and basically my entire phd was dedicated to implement these this idea which iber was talking about uh like implementing the variational reactive message passing and this presentation i just want to show you what you can actually do uh using this theory under the hood uh okay so basically in order to automate active inference right we need to automate bayesian inference and we have uh we have already a lot of solutions for that uh such as stan pyron umpire which is funded by google infer.net is funded by microsoft turing is in jula ipimc and many many many many so uh and basically these solutions are really really good so uh and they're really good at prototyping as well but our goal uh is eventually to be able to deploy this kind of systems not just prototype and we are really focusing on this particular properties for this automated bayesian inference uh so it must be uh low power adaptive real-time scalable it also must be user-friendly at the end and it must support a large scope of models if you want it to be useful uh so and in bias lab we we want to build such a software uh with such nice properties uh and it's always about tradels right so we do something better in one particular domain and maybe other software libraries they might be better in a different domain so but we are really focusing on this particular properties uh and so yes i will reiterate a little bit there's presentation so how do we achieve this so we have imagine we have an environment and we have an agent uh and the agent allowed to take some actions and the agent basically what he needs is to come up with some sort of good enough probabilistic model of its environment uh in order to do patient inference uh and in our framework we encode the model as a factor graph which not only models the observations but also actions and desired future uh and this this approach allows us to decompose the this complex relationships between variables and hidden states into some kind of structure and local blocks uh and it's not a block it's not a black box anymore so uh and the model itself may have some sort of background motivation interpretation it may encode your prior knowledge about some particular physical system uh and the locality of these blocks basically allows you to scale to millions of variables and hidden states it allows you to pre-optimize it maybe or maybe use like some

sort of different approximation strategies in different places uh so it allows a lot of very nice properties as well and we use reactive message passing to run uh actual variational bayesian inference uh it uses reactive programming under the hood to minimize the approximation to the variational free energy and yes as bered also mentioned it's it's it's very much related to actor model

uh uh and basically in rx and fur you can think of different nodes as actors themselves uh so and they have basically one single purpose is to send a variational message that minimizes free energy uh this is a very short and very high level description but it is essentially what is happening under the hood so we are not treating different agents which interact with each other as actors but we also treat the the actual components of the underlying model as actors themselves it's like a very hierarchical structure so this is the main central idea of of this inference uh so here for example a first example we can do an inference in a dynamical system and this example which is quite old already i think it's like two years ago so we we track a position of the object given some noisy measurements which are indicated by green dots uh the actual uh real signal we cannot observe it but we just can plot it uh is shown as blue uh and the inferred signal is shown as red and the data set is infinite the inference and just reacts on it and does not assume on any particular data size simply reacts in your observation as soon as possible uh yeah i'm actually not sure how smoothly zoom shares my screen uh maybe you can see it's a bit lagging in the animations i'm not sure because maybe zoom does not share it on a full frame rate um and also on the right hand side you can see how we define models uh in our framework we use julia as a programming language uh and so basically this is everything that you need to define this particular model and run inference on a data set and actually i like literally spend more days to plot it instead of inference right so inference was was an easy easiest part from watching was way much harder to relate to user friendliness uh so and we actually have plans to improve our model specification language make it even easier so now for technical reasons we have some auxiliary statements in the model specification language but we we are working to improve uh that and make it even easier this is another example which is similar to the previous one that uses much more complex and linear dynamical system of the double pendulum and the system is chaotic and

we can observe only a small part of it a lot of noise also indicated as a green dots uh and nevertheless given uh given good enough model uh you can infer uh the other hidden states with pretty much high precision and the code needed for that is also relatively short uh and yeah we also have uh examples with active inference agents uh that interact with their environment uh so the left uh uh left up shows uh yeah mountain car problem very famous problem the left bottom side shows an active inference agent which tries to control the inverted pendulum from falling in a windy conditions it reacts in wind we also have a demo of an agent that controls a pendulum in an ever-changing environment so on the right side you see a pendulum with an engine and legend has engine has limited power uh and the agent itself needs to reach the goal and the goal is uh indicated as a red circle so and basically in this demo we can change the environment in real time and see how the agent reacts so you can change the mass of the pendulum on its length or the amount of noise uh in the measurements or we can change the goal we can change maximum engine

power

etc right so the agent will still try to infer the best possible course of actions in order to reach its goal and it just never stops reacting um it's it's also actually possible to restrict engine power such that it will no longer possible to reach the goal right but the agent will still try

uh we have other cool demos with smart navigation and collision avoidance uh which are still under active research and the code for them is not available publicly it will be soon available but for example in the uh example uh we can define a set of agents with their boundaries and a set of their destinations so and we can see how they try to resolve their routes all together

um and we can have some static obstacles in the map we can see how agents can find their most optimal path

uh in order to reach their goals uh and avoid uh any possible collision and it's also not necessary to have static obstacles uh the obstacles themselves may move so on this demo we have hundreds of agents that

navigate through a map of obstacles that move from bottom to top the circles are obstacles and agents are depicted as uh depicted as small dots and they need to go from left to right basically avoid any sort of pollution

uh and as i also mentioned we want to perform efficient and real-time inference but we also do it like low power low performance uh on low performance devices such as raspberry pi or cool pi as an example

and we have some results of successfully running uh the bayesian audio source separation for example on cool pi

so it is actually possible uh we also run active we try to run active inference agents uh also on cool pi

uh so as the aforementioned inverted pendulum uh and as i mentioned we we also need to have a large model scope and basically rx infer has not been designed to solve any of the aforementioned problems specifically uh we have a large set of different examples in our repository uh different models different data different inference constraints uh we have examples for linear regression hidden mark of model autoregressive model hierarch models initial models gaussian process and so on so right so this approach is very versatile and for example if you um if you compare it with sort of a conventional software libraries where you let's say have a library that solves a common filter might be a very great library maybe super fast have top performance works great and very reliable uh super good but then you are constrained by by this particular model common filtering right and you can't really change it much uh in rx infer we we are free to define our own models uh which we can pretty much easily define a model that

essentially would act equivalently to common filtering equation and so basically in the demo that i showed before about object tracking it was essentially a common filter but was written in a in a probabilistic model uh so yeah that was my small uh addition to this presentation so our software is free so i'm mit license and it's open source available on github uh yeah and we would be happy thanks to be able to present where we would be happy to answer any questions thanks

awesome awesome awesome all right i'll just ask a quick question in the chat marco asks sorry if i missed

it are the collision avoidance demos real-time adapting to other agents behavior or is it a collectively pre-computed path um so basically they are not uh super real-time they're kind of fast compute this path like maybe five seconds or so but we are basically working to make it real-time so we know what is the problem we know uh where to improve and we will make it feel that yes almost almost next question do you have some comparative data with other methods and just more generally what kinds of benchmarks or what when you're talking with industry in different settings what are people like looking for that killer app of active inference or what are they looking for their key measures uh so i personally have a big paper about comparison with uh sampling based methods like hmc and also in my phd thesis there will be a comparison with nuts also other sampling based methods uh so long story short sampling based methods cannot really run uh this kind of sophisticated inference in real time uh they're very time consuming they do not really scale well to large problems which is really needed for active inference agents because if you have like large environment very complicated you will have you will have a lot of unknown variables in your model uh so yeah so there is a paper that compares uh data and basically we show that yeah our approach scales much much better so i personally run on just just a regular macbook uh laptop i run the model with two million uh unknown variables yes and it was like quite quite fast and then and if sampling based methods you may you may find yourself in the model with like 100 variables and then you wait like two hours or something and then it turns out that your chain did not converge so uh or something like that

oh yeah it's um people commenting in the chat like how far message passing and factor graphs have come and so to to bias lab and to bert at all we definitely appreciate this exciting line of research i mean there's so much to learn there and sometimes looking at the equations it can seem like it's like written in stone and just sort of the beginning and the end is you know variational free energy but then in your presentations you're really showing like no we we are hands-on that's where we get the interpretability the modularity that's where it really is implemented and it's like an information logistics challenge it's not like an esoteric philosophy question at that point no no indeed and i mean i i should say it's uh it's taken us i mean we are no geniuses right so our lab exists more than eight years and and you see all the people in the lab it's taken us many many many years with lots of wrong directions to get this to work to where it's now so it's a very long path but the the at this moment i'm pretty confident that at some point in the future and i don't want to put a uh i don't want to say in three months or one year but it we will be able to write a toolbox that will allow people to design a generative model and just press a button and and forget about the inference you don't have to worry about the inference anymore it will be fast and automated and that will happen and it will happen within a few years and uh and maybe somebody else will write an even better toolbox but i'm pretty confident that even our our toolbox um will be able to do that so um and i think that yeah you know people talk about uh so why don't we have the success of deep learning and generative ai right well they they have the success because of big data availability of big data big computers and toolboxes tensorflow and all the successes we don't need big data because agents collect their own data on in the field we don't need big computers active interest agents you know they they manage their power resources but we need a

really good toolbox because programming an active inference agent programming the inference by hand is just not doable um so we need a really good toolbox that really automates this we hope uh our xinfer will be one of the first toolboxes to do that i am sure that other people uh will also be working on it and and better toolboxes will come about um but i think the the optimistic message is that it will happen right uh and and once we have a toolbox like that then we can actually a large community can start building agents and we can actually show uh deployable agents in the field right that they work and they work better than every enforcement loading agent or whatever is out there right um so that's i think that's a very positive and hopeful message it's what we expect it's what we prefer yeah yeah yeah any last comments from either of you um from comments from us don't know we just i'm just very happy to uh well to get the opportunity and uh yeah i want to uh so yeah um everybody can download this toolbox um i think you at this moment you still should be a programmer to work with the toolbox and i hope you're friendly because you know it's not uh totally polished in the way that we want but it's coming right it's coming in the next years there will be a good toolbox for for almost everybody to use but people that are interested even people that are interested to work here at bias lab we have open position uh for phd students so we're happy to uh yeah to receive emails from people that are interested to have to work with us thank you dimitri anything in closing

no just thank you yeah again for our possibility to present super nice to be here

cool yeah well later in the year we will be discussing your two-part recent work and so we're going to be getting a lot into the details and i i hope that people in the institute and the ecosystem will be as excited as we all are so thank you thank you bye bye

all right

all right

for the next interval we are in the section hosted by the active inference institute on our recent paper active inference institute and the active inference ecosystem so two short messages first if you're in in the live chat i'll post the link right now to the paper and this can be more of like a conversational interval so if you have any questions about the um the ecosystem or the institute or um any of the stuff that we're talking about let's make it a conversation uh and then second message if there are any co-authors on this paper who are listening to this stream then message me on discord or post in the youtube live chat or just check your calendar and um i expect some others to join in as we go um welcome pablo we'll be hearing

in a little bit from you as a presenter is there anything that you want to just bring up or reflect on before i share the screen with the paper and we just walk through it explore it and hear questions

okay

um

i'll bring in the paper and again anyone watching live please just feel free to write any questions and uh

i'm just going to bring it so that pablo you're not going to see the paper

uh but it's it's on the live stream

all right

cool

well here we go this is a paper that had an awesome

co-authorship and even broader circles beyond the listed co-authors

of collaboration and contribution so just like everything that has happened and will happen

in our work it was massively participatory and a real learning by doing and a real enactment moment

the first version was released three days ago on saturday august 19th and i'll briefly just list all

the authors first we had the institute as the first author so that the citation can always remain like

institute at all institute comma active inference institute and we also had andrick gira john boyk

libra burian matthew brown rj corday scott david david s douglas pablo fernandez macchiere daniel friedman

holly grimm aval guanin carlou maria louisa annico virginia blue knight alexandra mikhailova

ali ramju adil razi yakub smiccault renan tamari dean tickles and alex viatkin so this includes

everyone from founders and officers and board of directors and formal participants on different

projects and different roles as volunteers or facilitators or interns and also people who are

continuing to join the fun and show up and makes the institute in our field what it is and what it

can be what we want it to be there's a table of contents for those who like to click through and

just broadly there's going to be a preamble on active inference but outside of the preamble this

paper is not conceptual it is not technical it is not philosophical although perhaps it can be read in

all of those ways but it's quite a logistical historical and vision oriented position document

after getting some of the active inferences out of the way in the preamble we're going to turn to

the institute itself in terms of our vision values principles and how we see moving through the

challenges that we're facing right now onto the specific history of the institute our current

organization model in terms of our governance and leadership structure as well as the institute

units the organizational units that support all the work we do we then turn out to the community and

ecosystem growth and development and go through some details of different roles that are stakeholders and

participants in our space and some of the tools that we use today for managing information and message

passing we talk a little bit about our communications plan internal and external both sides of the blanket

and point towards some of the types of support infrastructure and administration that we do and

that we will offer for the active inference ecosystem we frame that in terms of continual deployment

continual development and quality performance and growth evaluations across multiple scales

close with some discussion and then provide an appendix that overviews a variety of the open source

services that the institute has provided over the past so let's just jump in i'm actually going to

jump into the open source services first and overview them before we jump back up to the top

and look at the big picture because this is really to appreciate the work of so many contributors who

stepped up in the uncanny valley in the gray zone in the ecosystem as it's forming as it's always forming

and really made it happen and so many people we know are out there scribbling or taking notes or

printing out papers by friston at all or struggling and asking questions are there applications of this

how would this matter why would this matter is there a code version of this how do you connect the dots

between these equations like those are what we ask every day and so when people can combine their regimes of attention into a synthetic intelligence outcome then we go so far so the first category of works that the institute has developed are educational we've done video live streams since the end of 2020 till now more than 370 live streams all live streams are curated in terms of their transcript and we increasingly are developing automated and semi-automated methods for transcription translation sub captioning

language modeling and so on we've had textbook groups actually now as of now we have three completed cohorts of the active inference textbook group the 2022 textbook we have two courses that are being offered that are produced by external content developers and both of these have only started in the last three four months first is chris fields's physics as information processing and second is the work spearheaded by avil and co-organized with kairos research active inference in the social sciences both of these are like awesome and great learning experiences as we heard from earlier in the symposium we're working with sanjeev namjoshi on his incredible textbook although almost to call it a textbook is to either think more broadly about textbooks than one has or constrain what it really is that he's doing because it's such a unique real research development full-spectrum effort so i'm really looking forward to seeing how this plays out in 2024 until then for those who want to get involved with the testing that we do and we've developed the active inference ontology which is a critical resource for the ecosystem it enables us to center accessibility rigor and applicability across languages and settings by knowing what it is that we're talking about when we talk about active inference on the research side we've also developed the ontology in a research direction in terms of research on and about and with active inference itself as far as software goes our primary development first off has been the documentation and connecting the dots around some of these other incredible packages that others are developing for example just in this symposium we've heard from some of the core developers and open source contributors to pymdp the python package for generative modeling as well as just a couple of minutes ago from bert defrys and dmitry bagaev coming from the julia language rx and fur side and both of those are some of the leading packages for applying active inference so we do a lot of work to connect those dots and make it more accessible not everybody is going to jump in with all six feet to a programming language and what we've done with active blockference with the work of jacob smekal and others is to wrap generative models from active inference inside of the complex systems modeling framework catcat we've developed notation interoperability systems that actually let us pull back one layer from the implementational details themselves for example of rx infer or of pymdp so that we can bring it halfway to where the ontology and the visualization and the natural language exists and then use language models and other techniques to deploy it in an implementation

fashion we've developed a host of research publications we support people in all of their different stages of their career different seasons of life and some really creative work has been directly produced through an institute project or somebody who knows may have gotten good ideas or good feedback just by hanging out being in the game and we've performed literature meta-analyses to help grapple

with the scope and the quantity and the quality of what is unfolding before our eyes or before our antenna again even as we only so partially sample this ocean of information we're active in standard setting and continue to develop our qualification standards for active inference that could be the basis of education research and professional development we provide a variety of mentorship and facilitation service type activities including volunteer program internship program distributed facilitation and applied active inference symposia these activities just as all others that we're describing above are open source and they're also open to your participation so if you can follow some of those suggestions by jf time

boxing and prioritize and pay attention and be active all those fun things that he said and you're looking to do it in a semi-structured way with a volunteer program or in a more structured way with the internship then i heartily encourage people to get involved act now and in terms of governance activity

we have a scientific advisory board as well as a board of directors and together with the officers

those groups constitute the formal governance structure of the institute but there's also

so much more to it and we look forward to understanding what ecosystem governance can look like in the coming years

so welcome dean i just summarized the the work itself that was the pre preamble and now if you'd like to say hi you are pablo before we jump into more sections of paper or look at more questions that people have asked

how goes it or what was your experience of working with the paper or what brings you on to this strange stream at this opportune time uh basically why i'm here is because i uh for the first time even as as a acting board member with with the institute i was able to get a better sense of the kind of depth that we aspire to and i'm not sure whether or not we are going to be i i really don't know what directions we're going to be going in the next coming months weeks years etc but i have a better sense now as a result of participating in this development exercise uh of what the potential is and i probably had at any time and i've been with the active inference since two for a couple of years so that's saying something i think we had to get to this point to be able to write this document and now it's it's acting as a bit of a springboard now which way we go and how we land i'm not still certain about what that looks like but i think that's part of what the really interesting thing is um we seem to have moved away from the dock and now the real journey begins so that's my take on it so far moving away from the dock dock pushing off using it as a springboard continuing the adventure leaving the pre-print behind continuing to version it as a living document and also wayfinding outside of the dock the dock pablo

oh wait i don't hear you yet pablo

sorry oh i was mute hello um it was a great experience to go through the document and then

review it and reread it and thinking about it actually it's a living document which is super exciting and i

keep taking notes and and hopefully uh act on it on the future and uh and it's been a super uh experience to to find so many people with the same um some interest same interest and some values that i really think are going to be the basis of uh organizations on the future so i think we are doing something very cool

here cool all right well let's just walk through some of the sections in the 25 minutes that we have um again the paper is being shown on the stream but but dean and probably don't see it so we we kicked off with a preamble we wanted to introduce active inference as briefly as we could and point towards the fundamental first principles and physics-based nature of active inference and initially point towards its integrative capacity in principle and then in the second paragraph there's about 104 citations in this paragraph and that's where it's really only a sampling of the domains 21 domains here so it's a place where we can point to where is active inference being applied or how is active inference being applied or how is it different or better than another approach in a given setting setting which isn't to say that it does everything in all settings for all people at all times and all of that but this is to say that here's where one can look to find out more and even just building this and bringing it all together we we see so many opportunities for epistemic resources in the ecosystem and having real-time literature analyses and all these kinds of services that we want to provide starting on page five we turn to the institute itself and we write as of august 2023 the active inference institute is a registered non-profit organization tasked with identifying establishing and managing the sustainable implementation of administrative and governance functions to give components of the active inference ecosystem coherent forms and reliable channels of communication two publishing and licensing protocols that establish open and fair use and effective dissemination of community products within and beyond the ecosystem three services at the scale of individual humans and the community at large so that stability is protected while risk and uncertainty are minimized within the ecosystem and four organization and operation of cyber and cognitive security systems that ensure productivity inclusivity accessibility and safety and discourse in collaboration figure one gives a graphical representation just one of how we are now and for those who want to dig more check out the 2022 active inference textbook where you'll also find a golden ring and we kind of took that graphical motif and ran with it had a lot of fun and a lot of interpretations there so just here looking at figure one

dean or probably just want to add anything

yeah dean

so one thing about the building of this document um with a whole bunch of authors

uh it's often not made explicit but there are two aspects to learning about yourself one is to actually throw some stuff down on paper because you believe certain things need to be included and then the second part is the consolidation process and i think one of the most interesting consolidation processes was the first thing about the building of this graphic because i think what what it does is it tries to condense as as much as possible and yet still remain coherent around what the relational aspects that we are trying to make available are as opposed to this is what we've captured now and i think that this is what we've captured now look look look what our graphic designer came up with so if you've had a chance to kind of look at this and really analyze it and think about the layers and think about okay so what's the background here what's the foreground does this representation allow for a swapping can we have things that appear to be background or you know fundamental basis stuff

that could actually be things that we are addressing in the moment versus those things which we look at as particulars that in fact are generalities so again it's one of those things it's i i don't know daniel probably knows whether it's a where's waldo thing or you know some sort of uh artist that they came up with some strange twist on things but but the bottom line is spend a few minutes have a look because you'll want to go through your own building of what these relationships are and you'll also want to be able to consolidate if you i wasn't there when this representation was built but i do know that i want to be able to both look at it do a comparative analysis to what i think the institute should be and then be able to consolidate awesome yeah just a few notes on this and to give a visual overview since it may be clearer to see this then go through the prose that essentially describes it as much later on the golden rain represents the active inference ecosystem and i think we could talk about golden rings for a long time to come but coming in from the ecosystem on the bottom bottom for the reader not from the other side of the table are the projects and all the projects are participatory endeavors they're happening behind on and in front of the golden ring depending on how you look at it and that is the appendix and all of those different projects that we talked about and all of the iterations and new threads that i know that we're going to see in an hour and i know that we're going to see in a month and in a year and in a century and those projects are being supported through two units kind of like a department but an organizational unit one centered on education and one centered on research the education unit is called eduactive and the research unit is called re-inference so right there you kind of have active inference in case that wasn't clear coming from the institute scale supporting the unit scale are a variety of functions such as funding and partnerships communications and hierarchical processes such as the construction of enabling systems and at the institute scale other than administrative functions we have the officers the day-to-day operators of the work the board of directors which is the formal governing body and the scientific advisory board where we really want to call out broadly and have people who may have or may have not thought that they would find themselves in a scientific advisory board for example in 2024 this is your invitation and we really want to have a vibrant scientific advisory board who serves liaison type roles with different sectors and just gives periodic super valuable high leverage insight and then i guess depending on how you think about it as all things we have just a few terms they don't just represent bricks in the wall but it's more of like the the fibers we weave and some of these are already things that we do every day courses educational groups events live streams ontology research others of these are things that we know will be coming into place in the coming years and we know that the tapestry is just going to continue to expand so it's really exciting on page six we have another figure and this is just another way to represent some of the functions that we serve in the ecosystem this isn't the only or the final or a fixed representation of the functions that we have or the only way to slice and dice or any of that but just to connect some of the dots between the functions that we do and want to serve for the ecosystem so starting from the broadest let's just say where the listener is where you are right now is in the space of awareness and that happens through

communication some people who become aware through a search or language model or a video suggestion or

a friend or happenstance or synchronicity or however some fraction of the aware will pursue education and about education we have and will continue to produce the materials documentation ontology courses live streams and so on education is produced and rests upon a commons it's an epistemic commons it's a verified information environment and that is really the space the wildlife preserve as i know dean likes to refer to it not jokingly at all and underneath the commons is the support that does sit between a participatory commons and a formal governing council that support in the coming years may go many directions

but we know it's going to involve all of our favorite things like hardware software information behavior narrative and more and then we have the regular institute scale governance partnership sponsor and donor relationships such that we can first find a type of stability or sustainability and then more actively play a regenerative role in our ecosystem we want to dispense micro grants to people and we want to explore paid internships and be able to meet people where they are in their active inference journey and things have been happening to that end and through many means since 2021.

on page six pablo or dean want to add any thoughts

i'm super excited about the governance the education and the game and tools

in an hour we're gonna see the first active inference game this is very fun um

on seven we describe our vision our values and our principles and just to read the top level on the

values and principles they are active inference and exploration integrity and inclusivity dynamic

internal modeling anticipatory behavior and continuous development on page eight we talk about some

priorities and challenge areas these are challenges across multiple scales they're challenges that

individual people face in their journey to learn and apply in so many different backgrounds these are

challenges of national interest these are challenges of global emergency as well

these challenges are educational in nature for example related to scientific literacy and workforce development

and just general professionalization and changing times on the research side there are some

priorities and challenge areas related to the fundamentals of cognitive science and grounding the cognitive

science in physics getting that integrative first principles framework that we expect and prefer

about information science and diverse intelligences things are faster and different than they have been

an octave inference is perfectly poised for sense making and decision making in that space

the answer is perfectly clear and effective in that space user experience accessibility and socio-technical

design this is a major open question across different settings so it's not uniquely held to the institute

however we really do believe and really will commit to exploring new kinds of ergonomics and onboardings and

learning and experience so that we can come through and be how we want to be from the first time somebody

enters the game where we can highlight accessibility meeting people in the language that they prefer to

listen to and speak at the language in computer language or at the math level that reflects somebody's

preference in

that moment as well as being able to use augmented and synthetic intelligence systems to ramp up and even push the

frontier of what augmented people and other kinds of collective intelligence systems can do

cyber and cognitive security these also are broader topics

we've worked on these topics from an active inference perspective many have and we think that this is going to become increasingly relevant as things do become more and more challenging and uncertain in certain ways out there

scaling the active inference ecosystem we've heard about this from the narrow and technical side we had estimates of the

computational complexity of sophisticated inference about how many computer operations have to happen for a given

calculation to play out and that's sometimes called scaling active inference like okay it took one minute on one gigabyte of data

so how many minutes is it going to take on this computer with this much data

scaling the active inference ecosystem though is a broader question and that's where we

come to more of the social fabric and the real sense of belonging that we all want to have and co-create and provide for others in the ecosystem and so this kind of scaling is really transdisciplinary it's really skin in the game

it's as real as it gets and applying active inference some applications we've heard from today

some applications are cited in the early pages of this paper and more and more come through every day

and we support at the institute various applications of active inference dean what can you add in there

okay my mute is off good so here's here's a here's something that never came up in the writing of the

document but i think it's kind of interesting and i'm curious what you guys have to think about this

175 years ago an institution came and formed together it's called smithsonian institute now i don't know

if 175 years from this conversation today they'll be the equivalent of whatever the active inference institute

is 175 years forward but i think it's okay to start thinking in those terms it doesn't mean that we follow

in their footsteps necessarily but i think it means that we have a long-term view and that the things that

we write about ourselves now have some shelf life have some real sense of well we can't there are things that

we can and we cannot change but if you look at this if you look at the front page the landing page of

the smithsonian institute the first thing that they say is welcome which is what we say about the active

interference institute and then they say the smithsonian institution is the world's largest museum

education and research complex we are a community of learning and an opener of doors join us on a voyage of

discovery so in three sentences they kind of sum themselves up what's the formula for not just

existing for 175 years but being a force and i think again um denny whatever you decide to highlight in

the document today what i think we're really trying to do is say do the words that we put out now will they

continue to resonate with people for decades to come and again i don't want to sound all hand wavy but i

want to hold up an example of a place that really did set itself up to be something that people could return to again and again and again and i don't think there's anything wrong with aspiring to having that kind of place that kind of respect and i think that the things that you're pointing to right now hopefully set the set the the you know the we're preparing properly so that's what i'm that's what i'm taking away at this point we know about that preparation measurement cycle from chris fields don't we yeah

we go on to talk a little oh yeah please probably no no definitely i completely agree i've been thinking about that uh for a long time uh you you want to build and act uh for the best of the future and i think to reflect that on the paper it's super important so yeah definitely i have to do that yeah way ahead we described some paths that we want to take but and in the history of the institute we describe our origins but i actually wanted to come to to figure three and four on page 11 to speak to building one of the most amazing and encouraging things has been the all ages all time zone active build contribute mindset an earlier version of myself i might have thought like

hmm people who do variational bayesian methods and machine learning they're going to have no issue onboarding to active inference it's just going to be like as simple as saying yeah it's a variational bayesian model with action so just signal processing on the inbound it's just control theory on the outbound but if you're familiar with variational bays and variational free energy this isn't changing it that much oh yeah we use a particular partition and it's related to the free energy principle but don't worry too much you'll get there i thought that that would be the

path for so so many and future analyses will reveal what they do or don't reveal but what we found in our our enacted work as it really played out that there was inclusion and unique contributions to conversations ranging from philosophy of science to fundamentals of statistics and meta science and the role of quantitative studies and social systems all these amazing topics and people who just scampered so rapidly through what could be seen as prerequisite from a disciplinary perspective was beat was facilitated by the recognition that one was already in the space and in the way that they needed to be

to be an active inference and that was not written out before and even now the traces are barely recorded but it

will become to be understood more fully

and to your point daniel um i think especially what the pie chart shows is the actual spectrum aspect of this so i know that right now that that maybe the neuroscience and the psychology piece take up a bigger portion of this is the question but i think one of the things that we're trying to promote is the idea that um you can come from just about any background as opposed to oh all all paths lead to like as you said an earlier version of you said oh okay well i can see where everything is gonna where the confluence is going to no i think what this district demonstrates is that it's actually becoming more distributed the longer we we continue to to exist yeah this figure of course i'm super happy with everything that we were able to do but this figure is like almost a little bit like a um like a paper target that we're just going to blow out later because it was a fraction of only people who had formal affiliations listed which means only people for whom

we

had done full literature based polling of their information so it doesn't even represent the actual participation profile

this is more like the invitation list for live streams so we are going to see another long a long tail that covers the part that's not the long tail i believe

um the rest of the paper continues along possibilities challenges next steps more about the institute organization

various other pieces of information that potentially eager stakeholders or participants or donors or liaisons or other people would like to learn more about it's there to engage with we want people to hop into our discord and continue the conversation or email us or otherwise get involved it's why we we called this symposium enacting ecosystems of shared intelligence at the of course relevant suggestion of carl friston because we really do want to enact it so if you feel like you're on the sideline waiting to jump in or you're on the beach and people are out there surfing the 100-foot wave then maybe it's time to get more involved

so i will leave the closing word first to dean then pablo and then we'll go to raf's presentation

well just real quick again if you have if you want a consolidated document that speaks to the now

but also points to a direction that we none of us really know is is certain but is also really exciting

have a look at the doc pablo um yeah i'm here uh to make sure that i act as one of those people of all

backgrounds and perspectives that has taken advantage through one year of learning and acting uh on this

institute and hopefully see you in one hour so thank you and have fun thanks awesome thank you thank you

dean thank you pablo welcome raf also i'll throw on the paper so how are you doing i'm good what about you

uh pretty good other than this brief interval of white light i'm just hanging out

fantastic so i'm trying to figure out here how i can uh i'm using a different uh slide software so i need

to uh make sure that i can pull this up in the right way so give me just a quick second yeah great

cool so just just to um read off what's coming while you figure that um out so right now we have

raf kaufman and this presentation is going to be gaia an active powered network for planetary scale sense

making following this 30 minute presentation we'll have fl guinnon carlou on embedded normativity

that looks perfect that's perfect awesome all right thanks raf to you thank you all right i was

going to try to see if i could blow this up to full screen but i i can't i got it i got it on the stream

it looks good on the stream go for it cool so i'm just pulling up my notes here

all right so uh let me just test that this moves

uh did i just move the slide

great cool all right so uh as i said uh hi everyone it's a huge pleasure to be here and

thank you so much to daniel and everybody else at the institute for organizing this i'm going to talk

about what i think is the single most important application of active entrance not to as a detriment

to the others but it's sort of upstream of everything else and that is how do the 8 billion of

us humans make sense of what's going on in our planet and all scales and how do we use that

understanding to survive as a species so this presentation here is a bit of a technical

presentation and it has a lot of content it's going to move pretty fast and assume that you're familiar with some concepts but i'm going to try to leave time for q a at the end so first of all i'm going to talk about the motivation behind what we're doing uh in particular the concepts of the meta crisis and the third attractor then i'll give an overview of our network architecture and the application so far and cover the many remaining challenges and last but not least i plan to convince you that this is the single most important thing you could ever be working on and i'm doing this because we need your help so that's the taster of what we have and so as i said let's jump right into it with the motivation and what greater motivation could there be um if you haven't heard of the meta crisis uh it's a term that it's originally from dr who but it was borrowed by a thinker called daniel schmackenberger to talk about the total

risk for humanity and the biosphere that is posed by the combination of three factors the first being the increasing interconnectedness of everything from climate to food security to national security to biodiversity and everything in between the second being the the risks associated with self-evolving technology including runaway artificial intelligence and the third being the increasing uh omnipresence of coordination failures personified as the the god of coordination failures moloch and so the combination of these three factors as hypothesized by schmackenberger who leads to either one of two attractors one of them

being chaotic breakdown of uh all structure and the second being a reversal to oppressive authoritarian control and obviously both of those qualify as bad in my book i assume in yours as well

so the question of course is can we actually find a third attractor that is positive sum as opposed to negative sum and rather the question is can we design it so what i claim and what we as the the guy consortium claim is that yes we can uh and we can design this attractor and it's the design goal is basically to build resilience and uh stabilization into our bio social economic system and to do this subject to multi-level system constraints like the planetary boundaries uh desirability meaning it has to be something that people actually want so they will help it happen uh so it needs to preserve standards of living and so on and it needs to be feasible it needs to be achievable from our current initial conditions and of course there's a lot of design complicators in the mix uh some of which uh partial observation there's lots of information asymmetry going on uh there's a lot of uncertainty about how the pieces of the world system work from a scientific perspective there's obvious computability constraints associated to this and last but not least uh you can't really control what happens in most cases we can only create incentives recommendations and nudges so uh easy right of course it's not easy but we think we can do it if we apply

the the sum of the principles from cybernetics that we already know and apply them in a decentralized positive sum composable way so uh what we're talking about here is a decentralized network a decentralized hybrid

ai human network for planetary scale decision support and automation uh it features uh it recognizes that we will have human and artificial agents interacting by the billions in an open network it uh its goal is to facilitate both resilience or model alignment and coherence or goal alignment at multiple levels or scales

up to the global scale meaning the survival of the system as a whole

uh it will need to feature built-in incentives and governance and it will need to be privacy preserving intrinsically so uh the below listed participants are participants founders of the of the consortium i'm going to talk about it later but so this is the framing this is the motivation for what we're doing and uh and i'm gonna give a whirlwind tour of the architecture so far and i'm going to cover mostly what uh we've built uh the the version of this architecture that we've built so far uh thinking about this explicitly and then i'm going to talk about how this connects to other architectures that already exist in the world just very briefly so we've been uh framing this as uh an architecture for building decentralized digital twins uh which are local models of real world systems that run in network uh and the the role that such the show twins play is that they help us understand the costs and benefits of strategies and projects that happen in the real world and we in in our organization digital gaia we started out caring primarily about agroecology so how do you actually make better decisions uh meaning better recommendations negotiations valuations investments into a farm into a forest into a uh uh bio landscape to make it uh better to make it more likely to to survive and thrive so uh as a practical implementation uh it needs to tackle several real world challenges first of all uh it needs to handle distributed non-iid data uh data is sparse it's heterogeneous it's private it's potentially unreliable uh it needs to enable localized collaborative forecasting and planning personalized planning and privacy preserving it needs to uh accept the fact that it won't be able to directly control but only recommended nudge as we discussed uh and the the outline of what we came up with is this concept of the guy network which is a mesh of ai agents called natural entities or ants for for that act as proxies for real world systems each and runs on an engine called tangor which it uses to inquire learn plan and allocate resources and ends communicate in a language called the guy protocol they they use this language to independently interact with their environments and each other and if we do this right i'll show a bit more of this later but if we do this right the guy networks behavior approximates a single composite agent that handles uh that interacts with and couples with its global effect cool so um just to give you an overview of how this works behind under the hood the fangorn engine is really the heart of our architecture and the core loop uh that is mentioned here is a typical active inference loop but uh to make this work in very heterogeneous and distributed context there's lots of work involved so some just some highlights that i think are going to be particularly interesting for this crowd uh first of all so as i said this first the same core engine needs to be able to work for very heterogeneous context so each agent has access to a library of generative models and these models are specified using a python based uh program probabilistic programming dsl and they declare what context they're appropriate for using a shared ontology uh using that ontology uh fangorn can then select the most appropriate model to use online given the configurations like project configurations and the kinds of observational data available so using the model using model that's appropriate for uh for a farm or for a forest what kind of farm what kind of forest what kind of ecosystem what kind of of um you know their context energy project um i'm gonna get into the examples right now because there's more more on the applications

later on so uh we end up with a network of heterogeneous agents but that's okay they actually form a a single model in a formal sense because they're all connected hierarchically uh by a system of of hyperpriors and uh this means that fangorn needs to support inference of generic hierarchical generative models and we actually do this automatically using approximate posteriors with short flow constraints and so we have this global model that connects all the nodes and describes general facts that apply over multiple contexts and this means the whole network needs to jointly perform inference over this global model using a peer-to-peer or federated algorithm so initially this way this works initially uh the network sets the global posterior to be the same as the global prior each node then independently performs local optimization of the variational free energy and then passes the parameters of the new global solution to the peers in the network where they use it as a hyper prior and this works uh over a broad range of topologies and under pretty general constraints and by the way i'd love to i'd like to mention that uh there's a lot of overlap and interaction between the stuff that i'm talking about here and some of the other presentations that you're seeing today and i i wasn't able to reference them explicitly in this talk but i i'm sure you guys are going to see the connections cool so um what it turns out is that the shared parameters that represent our global model are effectively stored by the network stored in a federated parameter store meaning that each node stores their own local parameters their private stuff about their own local context and multiple nodes stored copies of the global parameters and this architecture because it supports heterogeneous uh local models it actually allows us to introduce another kind of nodes which we call ant mood nodes these are nodes that aggregate scientific and empirical information using a meta-analysis model and we actually are able to use that meta-analysis model to constraint or constrain our project level models or ants using quote-unquote imported knowledge that wasn't directly observed by them which is left to right in this picture but this actually also goes two ways so empirical findings from the projects uh what actually works on the ground for this particular project also informs the end moods posteriors which is a right to left and so we actually have a complete uh cycle of uh science and empirics all implemented using hierarchical active entrance so finally uh just uh among the global parameters that network learns by federated inference are the precisions associated with data coming from different data providers and this gives each node a way to consume data without trusted ground truth and basically to learn together which sources they can rely on for what

uh this is just uh some of the more core features we're working on integrating many other capabilities into this architecture not going to go through all of this uh but basically what's what we're finding is that we have all the building blocks to turn this into a full feature decision network complete with its own internal knowledge economy meaning that contributors uh contributions get attributed and rewarded in proportion to the to the free energy reduction that they afford cool so uh jumping into applications uh as i mentioned our main focus as as an organization has been on modeling agroecological projects like regenerative

agriculture and agroforestry and this exercises all the constraints that we talked about so far

so uh again i'm not going to go into full detail but to give an overview of what is the anatomy of uh

genetic model for a farmer forest so we start at the bottom right with observables that are things that we can measure about that ecosystem about that local context like the plant count and size vegetation indices from satellite data and so on these are linked to latent states which form a non-linear dynamical system that's parameterized by uh the policy the the things that modulate the the latent states uh like agricultural interventions and practices you can think of the timing of planting and harvesting what kinds of crops or trees get planted what products like pesticides get applied or not and more uh as well as covariates so external events like weather physical risks wildfire drought and so on um and so there's uh some scientific and uh measurement parameters uh that that go into this as well things like uh the effect sizes of the various actions and covariates and these local parameters are conditionally dependent on a set of global global parameters which as we discussed they they link information between different ends and different sources of knowledge um uh and uh so the same global parameters are used in the meta analysis model and they actually establish the link between what happened what is found to happen on the ground and what is reported in the literature um and uh so i'm not going to go into this because i i think i'm gonna uh run a bit long but basically the concept is is that we have these project nodes that correspond to to many forms and they're each on independent nodes in the network that are also connected to this common meta analysis model and uh together all of these uh together these nodes

they incorporate uh academic studies expert knowledge and uh locally relevant context they do so privately so only processing the local data locally and uh through this uh parameter sharing they actually jointly and iteratively estimate the the global posterior uh in a cycle forever for as long as there is new data about the project as long as there's new studies and new expert knowledge so so all this is happening basically in an in an online way and again the topology can vary but the principles uh remain the same so just to to see you show a little bit about how uh this works this is just two epochs of the the the posterior global posteriors and this is just location and scale for for two uh parameters uh the slope and the intercept of the relationship between uh uh vegetation index and the tree biomass in a one given uh hectare of forest and as you can see this this converges pretty pretty quickly it's comparable to two local local only methods um and just to highlight how this can be used to to also the limits of how this can be used to estimate uh source reliability uh here's a really successful example at the left it's pretty pretty straightforward to use this to estimate reliability of uh of uh satellite data and for other cases like very local very uh context specific data you actually need more data or more data from more independent sources that we had in this demo here um another very useful use case of this is as we discussed we don't have direct control over what happens at a farm we can only recommend and nudge often we also like we get there's a node gets stood up and it gets stood up for an existing project that has already happened and we want to find out uh what actions were taken in the past and the the same uh the same general model can be used to estimate uh what happened in the past and this example it infers uh when planting started and when harvests started for for different fields um so uh bringing it all together one of the applications that we've been using we're working on uh it actually uses all this machinery together to validate claims about projects performance and

[illegible]

[illegible]

this needs to be uh an enabler this needs to meet people where they are and that we need to be cognizant that change is not gonna happen overnight at least not right now right typically these things take uh take a while to mature and then they they snap into place and we're already finding some of this happening in some of the domains that we're working on where uh nothing happened for a very long time and then bam hockey stick and we believe that the same uh can happen in uh in pretty much anywhere so we're very optimistic great what are currently the greatest challenges for gaia as an organization i love that kind of ambiguous question like is that the guy consortium as an organization or is it guy but let's let's hear both what are the greatest challenges or frictions for the consortium and for the bigger picture too also uh i i'm gonna i'm not gonna talk about the the technical stuff for change i'm gonna talk about uh the the more meta stuff it's actually very related to what uh your uh previous presentation daniel about that paper uh about the institute talked about and it's really related to to uh building out this common what we were calling the common cognitive kernel for for sense making and decision making and uh relatedly uh breaking through this um uh this uh culture of sort of suspicion and uh surf wars and getting uh you know state people trying to stake their their claim to something so we are and always have been uh extremely open but it's it's kind of hard anyway to build trust so we are we're uh aware that it's going to take a while to build to build this trust with everybody and uh and i think the only answer to that is really to be as participatory and uh and transparent as possible also uh being humble and uh knowing where where we haven't we don't have the answer and uh uh asking people to to help us

make this um make this um make this um make this um make this um make this um

make this um make this um make this um make this um make this um make this um

a lot more you're going to hear a lot more about this consortium as i said it was just formed they just put up a website in time for this presentation by the way uh thanks so much to to marie and uh and of course steve and everybody else who's been a part of forming and uh and getting this to getting this ball to start rolling um we are um you know very early stage very formative and still uh very pliant so uh we definitely are we would love i i just gave you like my perspective on on a bunch of these things the the the shape of it that that that has come from from or work on this so far and even not all of it just piece that we could cover in 20 minutes or so uh but we are we're very plural we want to uh engage with you on uh on different paths forward different opportunities to to to build things to experiment to connect the dots to learn together and uh ultimately like we we really just want to uh provide a platform a convening space a protocol uh uh in the classic sense of the world a word of a you know a shared language a shared means of communication a shared means of understanding to find together what we need to do to to actually build this third attractor awesome call gaia trim tab thank you exactly thanks so much daniel and see you guys soon bye bye

all right

what a cool presentation next up welcome avell we have avell conen carlou with embedded normativity
so thanks avell to you hello can you hear me yep
and can you see the slides um i only see the top left of the slide
you only see this the top left yeah just yeah try again thank you
okay about now yep looks great thank you okay good so i am uh avenida carlou i'm interested in a
multi-scale collective organization this is why i came to active inference to begin with
and the core question i've been asking is um basically when we look at uh human organization there is
creative evolution there is a change through time of the specific patterns that are enacted and this
is very hard to account for and i have been uh trying to pin down more and more specifically what
what exactly is going on and how to model it so this specific work is a work i wrote in the context of
what my phd of course which is uh funded by the escape project a project in a cognitive archaeology
that's interesting interested sorry in how uh the culture the archaeological landscape shape cognition and vice
versa
the title of it is uh on a method no activity an active inference of agency an active inference account
of agency being flesh because you will see very quickly why so first we will go through the existing
work on uh and culture cognition in the context of active inference then we'll try to expose the
concept of embedded maturity which we introduce here and uh what it means and then we will look at the
more uh semiotic question around the externalization and utilization of normativity in uh cultural landscapes
so first in culture cognition in active inference so we know active inference i will go quickly over it
i still need to do it because there are onlookers that are not familiar so active inference say that
uh agents continuously predicts the specific cues sensory cues they will see and they predict the action or
policy they will not take
and they basically enact an ongoing stream of expectation that they reconstruct given you know a coming
signal
so this is best according for uh in this schema which shows uh the agent as a medical system which engage
through uh environment through action and cessation and all of these um specific
domains they minimize a specific function that is called free energy and whose minimization basically means
there is a productive process going on and a core property effective inference is that it dissolves agency as
we typically understand it we don't have agents that have like internal monologue that drives them to take
decision we have a conscious anticipation of what's going on and whatever an agent anticipate itself to do
it will do and so this goes for um computationally satisfying but very deflationary notion of what is going on
with agency
so the question then comes uh what is the relation between an agent and the cognitive landscape in which it
evolves
i will present very rapidly again the framework of skill and agility uh opinionated by brunberg kiverstein
and rydville and which says that skill agents which we understand as agent that understand what is going on
around them they will uh basically experience the world as a landscape of affordances which is opportunities
for actions
and they will their cognition pretty much reduces to the experience of this landscape and adaptive

engagement with it which basically maximize the grip the agent has on the system and essentially for reasons we will not go over because it's math the very uh existence experience of this landscape it is equivalent to the existence of constraints of our attention the attention that the agent gives to thing which uh translates the emily expectation that are at least two time embodies bad the mode expectation that the agent enacts

and so this becomes very interesting when we you know cross it with the existence of enculturation in humans

because the basic experience we have of our audiences of what we can do it is shaped culturally it is shaped by

social interaction wherein we learn with other humans about what we do when what we can do and what we can do

and this means that the basic human experience it's thoroughly encultured it's embedded in what one shared called cultural landscape landscape of cultural influences and it is embedded in cultural and material

regimes of attention so why is to pay attention to things and this is a figure that comes i believe from this year on the thinking through the minds where it shows basically um how the action and station cycle when we have

uh several agents they build a command environments that has epistemic cues on how the world is like and we

and we are inculturated to pay attention to things in a certain manner that makes it so we perceive and enact this

this landscape and so the uncultured constraint of all the flow of attention because of the what we saw just before the deflationary account of agency it means that the very expensive landscape cultural landscape that is

collectively enacted it effectively integrates uh intentionality it effectively uh integrates normativity in a way that clashes strongly with classical account of both so given this state of the art uh let us look at what i mean specifically about embedded normativity so embedded normativity is a kind of normativity i think you have you have got it so normativity it means basically the criterions by which we judge things to be good and bad

in the context of biological system uh they will look uh for specific i don't know chemical concentration and they

will avoid other this is the phenomenon of chemotaxis this is pretty basic and this is a form of normativity but for humans uh we will typically think of you know laws cultural norms um culturally shared way to judge behavior and again at least in the context of flow and choice theory we specifically think there is an agent

he has access to information and we will think what should i do and he will judge uh rationally about what is good and what is bad based on a collection of values which are internal to the agent this is not how it works um what i call embodied normativity is by contrast regimes of normativity where it is the uh the norms and

value that guide the agent's behavior it is embedded in the material and cultural environment so basically

the agent projects a specific set of norms and value in the very experienced business environments and it will perceive the locus of normativity as the uh environment and not as a form of internal judgment

um let us look at what can exactly how embedded motivity work it will be more concrete so uh first we can materially constrain people to do things so for example there are roads you can go off roads but it's harder and you can't really do that if you're a chariot so when you have a specific pattern of roads which is constructed by cities you are pretty much constrained to follow the roads

materially and for this we will talk of structural embedding and there is another regimes of medicine activity where we build on environments that will basically cue you suggest you into uh adopting specific norms and specific values for example escape um thinks that vertical patterns in archaeological culture they are somehow embedded within uh what we'd call vertical norms so hierarchy and certification in such relationship and for this specific form of embedding we can talk of semantic embedding

um what is must be very clear is that we don't have structural embedding on one side and semantic embedding

you know there are basically two mechanism complementary mechanism in which uh normative sorry

numativity occurs if we look at the city so this is um impression of sumerian culture uh city states we have a material organization that prompts specific kind of behavior

so for example we have a bridge and this bridge allow you constrain you to go through the river in a specific way if you are on foot

but this material constraints it is uh perceivable so by seeing the bridge you will also see that you can navigate the world in this specific way so you don't have a separation between semantic and structural embedding

we just have two ways in which um constraints of our behavior that are embedded in the environment work and something that is a core to the account of embedded normativity is again that it is not internal but we need to go into more detail

about what we mean by it's not being internal because of course we're talking about agency agents constrain what engines do it's not outside uh what it is is um embelly normativity is a regime of normativity where uh the norms and value they stem from the experience of fordance for example if i see uh fire i will not think about hey could i put my hand in it and get my hand burn hmm i think it's really painful i won't do that if i even think about putting my hand in the fire i will immediately feel bad the normativity it's embedded in my experience of the world i feel like the fire or more specifically the possibility of putting my hand in the fire is intrinsically bad i do not feel there is a possibility that is neutral and then i judge it to be bad but uh of course this uh this judgment it does not occur outside even if i feel like it's about my affordances it emerges from the engagement i have with the material and cultural niche in which i find myself and if we want to um to entangle what this means precisely we will have to go into more detail into the what i call the externalization and utilization of many normativity so let us look at something very simple how normativity work for embodied agents so embodied agents they do things hopefully in a niche that is material and this niche will be marked by the activity of my agents and this marks they can then act as constraints either through structural or

cementing a bedding in the agents environment because they basically signify that in the past i wanted to do that maybe doing that is you know good for example uh we have centuries with i don't know if centuries uh english word but we have path if uh animals that have legs this is actually important work sorry walk on a specific path they will compact the earth and it will become improper to grow anything even grass and you will have a path that will emerge that are pretty neat it's pretty visible and i'll tell you hey not only this earth is compact and allow you to walk on it like without effort but also this has been widely judged to be um to be a good uh outlook outline for navigation in this specific environment so the very experience of a path basically invites you it affords you perceive the affordance of walking through the path and this is something that again can result from your own activity that can be written into the environment by your own activity and so this begs the question of how okay there exists embedded normativity but how do we embed normativity what is the activity of embedding activity like and the word i will use for that is externalization we perceive the world because we externalize specific causes of sensation into an imagined objective example world so when we see the path we see a path we do not see compact earth that we then compute to mean that other things have worked on it we just see hey there is a path

path and this is because we have externalized a specific cognitive kind which is the path and this is a argument that several people made here lena steiner which if i remember well um worked on the the production of devices that help blind people see things through i believe an array of arbitrary device and this uh on their shoulder so the core argument they make is that if you give people that they will start seeing things they will start perceiving things as being before them rather than having vibration on the shoulder

and then thinking about oh what does it mean and so the basic argument is that uh perception is just um picture of what cause we infer

what is the structure we infer of our environment given the existence of a specific coupling device that relates action to sensation and vice versa

and this uh critically relies on the internalization of specific we could say a lot of things but i will say priors specific priors knowledge perception so me seeing the path as a path relies on me having the prior knowledge that there is such a thing as path

that there is such a thing as path and i should be expecting to see them around

and so you have um duality uh of externalization and internalization in the production of embedded normativity

i can only uh see norms out there if i've internalized integrated a specific apparatus of perception that entails that there are indeed those specific norms that are embedded in those specific affordances

and so this uh produce a marginal question of uh semantics basically of information theory how uh do reading and

writing and reading about vt works uh so when i extend lies i treat the world as having specific natural categories

that i pretty much imagine and i can use them because i imagine them as an excellent memory that allow me to read

and write normative cues uh this can be um you know letters uh pure signs but as far as i'm concerned here all signs are at least implicitly normative there about how you should interpret the sign and uh this means that we can experience a specific landscape of forenses and we can we can write uh things out

there we can write normative cues by behavior that will then constrain further behavior but nothing tells us that we can um that we can communicate through this the communication it works uh because of the internalization of prayers if a given community of agents have internalized a specific set of prayers that is a set of prayers that enable externalization of normative cues then only it can use the environment as a way to um to write and read normative cues as a embedded normativity thing and this communication it can be read in terms of information theory of course it can be read in terms of semantics but what is uh critical to analyze that the process of internalization it is there there is no um trust system there is no private key that allow you to be sure that the way you internalize norms is very specifically the way in which the emeline maturity was encoded to begin with there is a loss or there is ambiguity or there is i'd rather say it like this an active demarch of reconstruction of given priors by the agent and so this gap between the intent of other communicants and my intent by understanding of what they meant

by their active engagement active engagement with the world it builds a tiny gap in information that is somehow enough to drive the open-ended evolution of cultural landscape and this is something that i do not think is embedded in active in france and i think is of major interest to research to the research at least in the evolution of uh open-ended evolution of cultural landscape so uh to sum up things uh i just presented a paper that is called an amazing motivity and that speaks unsurprisingly of embed of embedded normativity uh first major point is that if active inference is correct then um our notion of agency is dissolved essentially in the notion that we can perceive and enact cognitive landscape uh umwelt if you want and given uh that this is the case we will experience the norms and value that guide our behavior as a property of you know the cognitive landscape experience the affordance that we can perceive and this specific regime of normativity i call embedded normativity and the third thing is that you have a critical thing going on you have a semiotically constraining thing going on in the externalization internalization of regimes of embedded normativity because uh those two things they respectively enable normal emitting activity to exist to have a causal power and they enable also the open-ended evolution of the norms that are embedded in a given landscape in a given material landscape and therefore the open-ended evolution of the associated cultural or cognitive landscape so thank you i will uh unroll my references slides in case someone has the really needs to know what the reference is and is ready to move around the video for this and do you have questions awesome thank you avell while people are typing any questions could you maybe just give a little introduction to the active inference for the social sciences course like what brought you to want to facilitate this effort and where are we at and where do you want it to go what do you want it to be for people good point thank you uh the active inference institute organizes uh and kairos research organize a course on the on active inference in the social science that is uh biased essentially in explaining the um

basic demarche the basic hypothesis and the consequences of the cultural landscape formalism that i hinted at slightly in this presentation the goal of the course is to give a basic ontology that is in the interface between complexity science let us say physics written large uh cognitive science and social science to uh afford interdisciplinary collaboration in the um in the social science and in the study of the evolution of human societies and i think it is cool but i'm biased because i've noticed it so maybe you should ask daniel which is absolutely embarrassed whether the course is cool i i've been enjoying it a lot uh we just finished last week with my section on collective behavior first we had the introduction lecture from you then ben white on the basics of being an active agent and we have lorena and mouse sections upcoming as well as your more dedicated session yeah it's been a really fun uh experience and it's going to provide a really useful missing piece in the ecosystem i guess just one question i have how do we take some of these topics you're bringing in about the embedded normativities and think about them digitally like are we talking about conceptual embeddedness in information environments and or are we talking about the ergonomics of sitting at our desk or how do we bring that kind of pleasure path stigmergy on the ground into the digital are we talking about something that's embodied and embedded in the information space or are we still talking about bodies

so if we talk about humans we talk about bodies with the brain it's that do things in a material environment but your question is uh i'm not sure understood is how can we bring this specific concept in the study of digital phenomena like twitter internet yeah is that it yeah okay so you have basically the whole study on the uh affordances you have a so the discipline of design has incorporated a while ago the national affordance in its structure and most of the study in the digital design center on what platforms afford or do not afford and you have a richer stance when you take into account not what you can do with a specific interface but what the specific interface invites you to do given a set of cultural written knowledge it can be self-learning but given a set of power that you share with with the um platform in a way that is uh formal but indirect and um

so yes i would look at it in terms of in user interface design i think this concept specifically is useful for um for interface design uh then you have implicitly the question of um the flow that platforms such as i don't know uh twitter facebook um how specific uh structure environments bring specific collective dynamics that might be a bit um richer not richer specifically but that might go in a direction that the user interface question do not go into and that can still be modeled in terms of constraints and therefore in terms of embedded normativity is that on point we'll go to a question yeah that's interesting about what the interface does and then what it invites it might invite things in order for that invitation to be fulfilled it has to actually provide that possibility but it might invite things that that aren't relationally possible like there might be a wall which invites climbing to a given observer yet they cannot climb it um and then on the other side is a capacity that isn't invited and that is the the adjacent possible of tool use where something that was being used for this purpose it's like all of a sudden it just fell out of someone's hand they grasped it a different way it invited a different engagement and then that can become re-entrenched and that points to that sequential like making and breaking of new spaces

that your work has featured over the last few years yes exactly this work has been developed with uh while thinking about tool use while thinking about cognitive archaeology we're thinking about the experience of cities the way urbanism uh from specific behavior uh but um if i can get back on the internet thing i'd like to make it more specific so the interface design as far as i know is focused on interface but is that is not surprising uh but in doing so it is uh focused on specific affordance and what you can do uh you have another layer in uh ecological psychology which is the layer of uh solicitation so for example twitter does something very specific which is when you log in it show you it shows you the number of identification you have then it removes it and you have to wait to know which were those certification and you have to wait no one second not one minute but a random amount of time so this is uh what they call dark patterns they are patterns in design that are explicitly designed to create addiction so this is a layer you can't afford in affordance because uh yeah from a performance perspective there is just a moment you can access the information and then a moment you can't and then the moment you can again there is nothing going on if you don't look at intrinsic motivation and the way a specific structure environment is perceived the semantics semantics maybe the meaning that uh given landscape has for an agent and even if you look at this you are still still looking at the platform in a way that is um is embedded from the specific patterns that specific patterns of things that people do on the platform if you have twitter without the aggressive qrt's and the um way to aggressively signal that you are in whatever it is is going on you don't have twitter much you have mastodon maybe that has very similar affordances and very very very dissimilar solicitation so this is this is why you have to take a multi-scale account of things and when you take this kind of things you have to look at the semiotics basically behavior how specific coordination occur at specific scales around specific uh communication specific message passing and once you look at how specific message become meaningful to specific agents you are you you have a problem basically because this is subjective this is constructed through the activity of the agent and this is uh whatever is the range of time you are looking at you will have the construction of new possibilities of adjacent possibles as you put it and yeah this is the this is the uh big question that i think we are missing at the moment in active inference

awesome well thank you for this presentation thanks for the social sciences work and we're all looking forward to seeing how the projects play out thank you goodbye farewell avil
all right

the next presentation

is with pablo fernandez maquetta so welcome back pablo hello

okay okay okay i did take us on an adventure of curiosity yes i'm gonna start sharing my screen let me know if everything is all right

should be perfect yep cool

so this is one of my favorite memes and um

and probably most of you know it uh this reference uh is um um the cave of platon and uh

to me it was kind of fine treat if uh truth that uh now with active inference seems like we are closer to

[illegible]

where it was the first at least what that i found uh place where it was and the license it was a creative

commons license uh of not commercial use but uh it it was an open license which i been uh a part of uh creative commons community for a very very very long time since i was uh uh working with musicians and i think is one of the best things that happened to the internet um this is these are some of the pragmatic uh uh things that uh the audience cares cares about hopefully we will have some of these on our games uh ai for sure is going to be there um free energy principle is going to be one of the things things that hopefully we help to minimize and to make it fun in our games or happier games or and games that

help organizations to be better uh the human factor uh it's super important for me and implementation and action i learned by doing and it's one of the values that the institute encourage more so uh i just jump on things and try to understand them through doing and sharing as i'm doing now uh here are the epistemics and um collective uh is one of them that um as well uh super interest um agents of course uh practical and curiosity which is the adventure that um uh i have the feeling that i'm being involved all my my life and that i encourage always my my two girls to to be curious and to uh open things and look behind the doors and just be an active here are some values that i like to point which are the culture of the culture of the culture of the culture of curiosity that i did mention a lot continuous learning uh because that experience on the school i think i i just keep going and keep going and it's something that sometimes um bad experiences could turn out in good habits uh learning by doing as i already told um integrity and inclusivity honesty accountability a big one um trying to be professional on everything that you do um for me i started as a human being curious um people of all backgrounds and perspectives are welcome to them to the institute um i could i would be very happy to be the link and to speak with uh uh someone that has doubts or if i couldn't help i think i could help on the very early stage and and if you're not very technical person i think i could help there and and yeah and as i we are learning by doing i would love to everyone uh that's uh that's on the stream uh i'm gonna share this link which is uh

a new a new game that we are testing and i'm excited to to share it with with you uh uh it's the adventure of curiosity and here daniel i'm gonna copy and paste the link uh it would be awesome okay there's a tiny typo on your slides it just ends with a curiosity thank you but i'm gonna i'm gonna paste i'm gonna paste it into the youtube live chat okay thank you uh yeah i told you i'm dyslexic i have all kind of titles around my life sorry for that i always try to improve it uh let's um okay so you shared uh yeah i put it in the youtube chat so people can head over there all right yeah could you tell us about this platform and this is the first active game yes um we uh started thinking on how to do a game a while ago and we decided to go with this um picture that explains how it works actually it's simple and you have uh four pieces the internal state external states and in the middle you have the sensory state and the active state so here is the mission you have for some instructions that uh you can read and would be fun if you go through them and you can go either way and here we are going to go to the sensory state where you're gonna listen to a music where some notes you can censor that are not in the famous song of for release

so you have to take notes

and you can go and you can go and you can see that we are on the mark of blanket which so i saw everything very delinean as dali but painted and here is your active uh part of the game where you have to do something which is

add the different notes that your uh professor music professor is telling you you don't miss because you probably on the uh

uh external uh state are trying to play the music so you have to infer that you are missing some notes and go

and then go to the active state and start uh walking through the right notes uh if i miss the note i just follow and fall into the nothing and come back into my internal state so uh probably would have to go back and sensor again

and once i take the right actions and the right notes i can go and solve the puzzle and the idea here is to have a tool a gaming tool to explain or so you can experience how this uh active inference and free energy works and once you

go to the end of that adventure which i'm not gonna finish because it's something that players should do you're gonna have a door where you have to put the sequence in low case of uh of what were the missing notes and it's gonna allow you to come inside the treasure room where you will have uh feedback uh uh feedback that you can send us and we will be very happy to have it and then uh if you have a wallet you will be able to have a digital asset if not we will add information to help people to go from web two to web three and to get those digital assets and here to be developed would be future adventures that i would be very happy to

do so this was pretty much the presentation and hopefully we have played new players in the coming days i hope you have fun and i really really encourage anyone to discover and play with all these ideas and ideas and thoughts and experiments and things that i have the gut feeling that are gonna revolutionize uh the world and how we can improve in it by reducing uncertainty so thank you and i don't know daniel if you have any comments or

something in this last 10 minutes oh yeah well we can definitely talk for a few minutes i'm i'm seeing many fun comments and questions in the chat so people can can post it yeah the gameplay is smoother on even a normal

computer it just like it may look choppy on the live stream but it's quite a nice um playing experience like maybe just a little context like how did you get into building these worlds or what kind of tools are these and we obviously don't see them in today's mainstream educational and research offerings but like what could that even be to bring together this kind of immersion and role playing with education research

yes um i uh uh we i i i i've been playing role games and role games uh you have uh very uh diverse kind of players so i i are playing are players so i i was playing with uh guys like me like uh have um had a struggle through the uh uh school and guys that uh end up being uh researchers like yourself because they were very good and and in those environments and and and and for me it it was like

we are sharing a lot of cognitive uh um values and information and they are really having fun with me so uh that what what the academia is telling me is not isn't it's not what i'm i'm i'm feeling in in the rear in in outside of academia so that that through the years uh it became more and more clear that was one of the drivers to have a very diverse kind of friends and and and that one of the most important things for me to learn it was having fun and to be a good professional i have to be enjoying what i'm doing and to enjoy what i do it has to have that gamify feeling of it and so in 2021 uh i i just went to just try to explore that road uh because uh blockchains what what can do is have a persistent sea of of objects and attribute attributions to those objects and to those wallets that you can add and build on top of them it's like having a huge uh database for a game purpose and and then you figure out that a game is a kind of our organization so you need rules you need to share values you need to uh have like uh what we are building now which is the session cedo which you agree with the players what's going to be about and how it's going to work and and we are uh uh building this um fantastic uh storytelling of uh how uh kind of world where it was in the uh early 20s compared to what's happening now in this uh imaginary world that's complementary to what's happening in real life and we are just exploring those areas for three years and once i got into the into the institute uh we started sharing these ideas you uh send us to do um um system thinking course that it was super uh cool which uh we we explore and learn and this uh document that i can say it was very uh important for us to understand and we uh fund find out the necessity to be able to computize what we are doing so we start focusing more on ontology on on on all the bricks that you need to understand, to build the skeleton of something that could last for decades, which is the idea.

And we are trying to build it so everyone can copy, paste, fork it, and do whatever they want with the game, adapt it to their own organizations, and use it on their favor.

And we are just exploring and having fun.

That's awesome. And there's so many pieces there.

It's true about the fun and the curiosity.

Those intrinsic motivations, like being six hours deep into counting ants in the desert or 5,000 pipette pushes deep or whatever it is in the trenches and in the last mile.

And it can be easy to lose sight.

But when we lose sight and when we put out the fire and the spark, it's a cold dark night and no career or status or extrinsic valuation is going to provide like simply the joy and the, the deeper meaning of like two people going on a little adventure running around this space, being inside active inference rather than active inference, being like this high speed train that's leaving anyone behind,

or you're on the outside and it's taking these twists and turns and you're, you're getting thrown off.

It's like, it is cool and air conditioned in this space.

And people play and watch a lot of games where they wield violent arms and not even like saying whether that's good or bad.

It just, it is a genre to want to have a camera either in a character's head, first person perspective, or a camera in a second or a third person perspective, which you can switch between here.

And like those kinds of topics about like egocentric versus allocentric navigation and our ability to have a perspective swap on ourself and be able to take that kind of like out of body, um, situational awareness in real spatial settings.

And just to be able to cruise through this space.

And it's like, yeah, there's no, there's no right way to boomerang from extra.

You can go from external to sensory.

You can go here or there.

It's, it's a, um, it's an appetizer and, uh, a fertilization that like, we've really just never seen.

Okay.

Yeah.

Thank you very much.

Daniel for this.

Any last thoughts or comments?

No, just enjoy while we are here, which is.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

or mission.

All right.

Thank you, Pablo. I'm sure everyone's going to really enjoy this, and let's also, of course, continue to develop these games and continue to offer these on-ramps and have times where we're meeting up in these spaces and playing around.

Thank you.

Cheers.

Farewell.

All right.

Welcome, Mal.

This will be the final of the presentation sessions in the symposium. This is Mal Alborosan with the presentation Shared Pretensions in Active Inference.

So, thank you, Mal, to you.

And you're the first you're the first to join with an AI steward

slash guardian.

Awesome.

Shall I share my screen?

Yes.

Let's do, let's do, let's do.

advance just a second.

I'm looking for a window.

I think this will do.

Okay.

So, can you see the screen?

Yep.

Looks good.

Thank you.

Awesome.

So,

unfortunately,

I imagine that you are seeing the entire screen and not just the presentation.

I apologize about that.

I got it.

It's cropped.

I got it.

Okay.

Wonderful.

Thank you.

All right.

So,

welcome to the presentation on Shared Pretension under Active Inference.

We're going to cover a lot of topics.

Shall I introduce myself a little bit?

Or have you already done that?

Awesome.

So, my name is Mal Alborosan.

I'm a PhD student in Cognitive Computing.

I work in Active Inference in the interaction between social sciences, philosophy, consciousness science, and artificial intelligence.

We're going to start a presentation on pretension, pretension, and active inference.

So, I'm going to define a lot of terms.

It's going to deal into philosophy terms that are very deep.

And so, let's get started.

If you will just give me one small second.

And we're back.

So, shared pretension is a term derived from phenomenology, and it refers to the shared anticipation of future events among a group of individuals.

This philosophical study of experience and consciousness becomes particularly interesting when we consider it in the context of active inference.

When we combine these two concepts, we get a powerful tool for understanding how groups of individuals can coordinate their actions and work towards a common goal.

This is the main focus of the discussion today.

So, imagine a group of people planning a trip.

They all share a common goal.

To have a successful and enjoyable trip.

This shared goal forms the basis of their shared pretension.

They all anticipate future events related to the trip, such as booking tickets, packing their bags, and visiting various attractions.

As they start to take actions towards their goal, they're also engaging in active inference.

They're making predictions about what will happen next, and then adjusting these predictions based on what actually happens.

So, for example, they might predict that a certain flight will be the cheapest based on their past experiences.

But when they actually check the prices, they find that another flight is cheaper.

So, they adjust their prediction and book the other flight.

This constant cycle of prediction and adjustment guided by their shared pretension helps the group coordinate their actions and make their trip a success.

So, we're going to delve deeper into the concepts and explore the role of past experiences known as retentions in shaping our current knowledge and future predictions.

Let's explain first the concept of retentions.

Retention is a key component of time consciousness that retains the past trajectory of a temporal object.

It's the preservation of the content of a now past hyletic datum in our present consciousness.

A hyletic datum refers to the raw, uninterpreted sensations that arise in our consciousness.

When we retain these data, we are essentially keeping a record of past sensations and experiences that can inform our present and our future actions.

In active inference, we can formalize retention by thinking of moments as binning of states, which are bound to a hierarchical state.

This way, retention is seen as the binning which becomes the current state in forming the next expected state.

To further connect retention with active inference, we can consider the way in which retention plays a role in updating an agent's belief and generative model.

The past experiences influence the current expectations and actions, thus shaping the agent's understanding of the world and the future.

So let's consider a practical example to illustrate this concept.

If you had a good experience at a particular restaurant, you might choose to go there again in the future.

Your past experience influences your future decision.

In Husserl and temporal phenomenology, we would think about much shorter timescale.

But if we extend this concept, we can see a much larger structure wherein the retention is simply the past state.

It's like having a mental note that says, this place is good and this is how an example of retention works in your everyday life.

They're not merely representation or memories, but presentations of a temporally extended present, which distinguishes them from recollections.

This involves the presentation of some already constituted content in conscious experience.

Retentions also play a crucial role in mapping past knowledge onto current beliefs, contributing to the development of shared narratives or goals that we will delve into in the rest of the presentation.

Retentions are also a crucial role in the future.

They can be former now points and the current awareness of the past and far sedimented retentions can be engraved in scripts or deep temporal priors or more fluid deontic cues.

This phenomenon corresponds to a specific application of message passing between agents in the present moment and in the sensorium.

This phenomenon is a crucial role in the present moment.

Retentions and protentions may also emerge jointly within a group, forming a collective understanding that transcends individual perspectives.

Protentions, as we saw, refer to the anticipation of future events based on past experiences and current knowledge.

It's thus forward-looking.

It's the forward-looking counterpart to retention, which is backward-looking.

Protentions is the anticipation of future trajectories of a temporal object.

It's the process of looking ahead, predicting what's to come based on what has been and what currently is.

So in active inference, it could be formalized by considering the future states that are expected based on the current state,

based on the current state and past states.

Protention is thus the anticipation guiding the next action.

You can consider the way in which protention plays a role in updating an agent's beliefs and generative models,

influencing the actions and thus shaping the agent's understanding of the world and the future.

So for instance, if you see dark clouds in the sky, you might anticipate that it's going to rain and decide to carry an umbrella.

This anticipation of future events is protention.

It's also going to determine whether or not you're wet when you go outside and therefore whether or not you

expect to be wet.

Your brain is constantly making predictions about what's going to happen next.

So let's put this in the context of temporal flow, which ties together those concepts of primal impressions, and we will see retention and protention.

Temporal flow is the structure that represents the continuous progression of time from the past through the present and into the future.

Agents are constantly revising their generative models, and they do this by integrating the observations, which we could refer to here as primal impressions.

This updating process is a balancing act between maintaining the predictive density of the agent's current beliefs and adapting to new information.

Pretensions and protentions structure time for a given agent, creating a framework that shapes their perception and anticipation.

Different agents or systems will exhibit varying qualities and quantities of retention and protention, which resembles something like Mike Levin's concept of cognitive light cones.

So given the presence of retention, we can anticipate a protentional horizon, not merely as an arbitrary expectation.

Repentions are thus the point at which agents' beliefs about the world are updated to match their current observations,

which leads to a more accurate understanding of the environment and allows for more effective goal-directed behavior.

Primal impressions are the fulfillment of a protention, which is the time-bound nexus of perception and existence.

It is the resolved probability that redefines the local gradient.

It's in primal impression that reality collapses.

The primal impression is thus modeled as the essential Bayesian cognitive moment.

It is the juxtaposition of prior beliefs or retentions with ongoing sensory information in an unfolding future-oriented present of retention.

So now we have a description of temporal thickness from phenomenology, which is compatible with generative modeling in active inference agents.

For Husserl, this flow of conscious experience is primarily made up of sensations that well up in a raw, uninterpreted form that we called earlier hyaluronic data.

So this data is the now phase of a temporal object.

It is formatted according to the eidetic or cognitive laws of consciousness.

This is specifically relative to inner time.

This is a way to understand how temporal flow structures, how agents are able to make choices, act to self-evidence, connect the passive structure with the active component of decision-making or goal pursuit.

[illegible]

Pretensional goals are future-oriented objectives that individuals or groups strive to achieve.

They're not just mere desires, but they're actively pursued through the act of will that we called earlier fiat.

The fulfillment of a pretension drives the agent towards a teleology, that trail set we discussed.

The fulfillment of a prediction entails less surprise, and therefore the agent self-evidences by seeking to accurately predict their surroundings.

Agents select the series of actions of policies, which generate transitions between states that produce expected observations.

So, through the choice of policy, an individual can reach their protended goal by minimizing the expected free energy that a given policy affords them.

This seems pretty central and simple in terms of active inference.

So, farther pretensions are represented by the policies of longer time depth.

In a hierarchical model, this corresponds to a policy tied to a higher level action in a slower time scale.

The choice of policies driven by the expected free energy of that higher order policy, or the entire multi-scale generative model, is based on the agent's preferences, allowing an agent to conduct goal-directed behaviors across temporal scales.

So, this is where we can cache out Jeff Yoshimi's trail sets, where agents assimilate actions into the anticipated continuity of objects in their surrounding world.

This dissimulation ensures a smoother understanding of the environment and helps agents adapt effectively.

Near pretensions act as the near-present contextual proximity, and the far pretensions rely on passive synthesis of deontic cues, which are sedimented retentions in the environment.

The preferences of the agents aligned by policies correspond to attracting sets, which in the socio-cognitive setting can be instantiated as narratives.

Having shared narrative is a necessary aspect of any exchange between artifacts, people, institutions that have some kind of attracting set.

Generalized synchrony, which we will discuss a little bit later as well.

So, a team working on a project has a potential goal to complete the project successfully.

The shared narrative might be the project plan that guides their actions.

It's like everyone in the team has a shared story about what they're trying to achieve and how they're going to do it.

The relationship between current states and future observations is captured within the B matrix transitions.

This matrix represents the probability of transitioning to a new state based on the current state.

The B matrix transition is utilized to determine which new observations is more likely to occur given the current state and potential transitions.

In the Husserlian framework, the expectation gradient structure plays a crucial role in defining the depth and fulfillment of pretensions.

The structure represents the varying degrees of certainty and confidence agents have in their predictions about future states.

The topology, thus, and the arrangement is interrelation, sorry, there is an interrelation between the pretensions or the arrangements of future expectations, which is proportional to the expectation gradient.

The higher the gradient, the more detailed and accurate the pretensions are likely to be.

The potential horizon represents thus the static states of the expected futures of the temporal object, while

the continuation horizon is the amalgamated trail of the temporal object.

Potential and continuation horizons are carried in protensions, while the imminent horizon consists of the primal impressions.

So agents can coordinate through co-construction, where agents or individuals interact and collaborate to build this shared understanding and shared goals.

This process is deeply interconnected within a complex web where agents typically co-exist with other agents or subjects.

formats, forming a collective often referred to as multi-agent groups, swarm ensembles, or assemblage. By aligning their beliefs to the signal found within their niche, they can better adapt to the environment and develop a shared understanding of their surroundings.

Sharing information with other agents is beneficial as it allows them to trust one another and pool their resources. This collaborative approach is preferable to a conflict-driven scenario in which every other agent is perceived as a potential source of risk and surprise.

By sharing information, agents can collectively construct a more accurate representation of their environment, leading to more effective decision-making and action. Through these interactions, shared narratives and goals can emerge from a co-construction of the world in each agent's generative model. So these narratives and goals are the result of a dynamic process that in fact can be mathematically modeled through the equations that we introduce in the paper that this presentation is based on, but that I will present a little bit later.

They capture the process of mutual learning where agents adapt their generative model based on the information gathered from their interactions with others.

Agents may start off with slightly or greatly divergent generative model. So in order to minimize surprise, there will be a natural alignment within a group as agents adapt to the evolving dynamics of their environment and social contexts. For example, again consider this team working on the project. Even if you have your own understanding of the model or of the project, you'll interact and share information and learn from each other and then adapt the model. Over time, you'll develop a shared understanding of the project and a shared goal for its completion. This is an instance of co-construction. It's like building a house together where everyone contributes to the final structure.

The emergence of shared narratives and the learning and recognition of them may be made more efficient through certain mechanisms such as language. In an experiment illustrated in a paper that Carl Rouson, the two agents share the same sensorium, ask each other questions. I think it's a paper on 20 questions in Emma Holmes, with a shared narrative entailing a common language. This illustrates the marked efficiency and the minimization of free energy simply by having a shared language, a common ground. So shared potentials are common future anticipations or goals that emerge among groups of individuals through co-construction. It allows them to better anticipate each other's actions and intentions and work towards a common goal. It's strongly tied to the conscious efforts of individuals to work towards that goal and to effectively create an organizational structure where we're engaged in a conversation, in an exchange, assuming that this exchange is meant to align our generative models.

So think of, for instance, the football team. Again, they have a shared pretension to win the match. They anticipate each other's moves and adapt their strategies based on the ongoing game. Everyone in the team has a shared vision of victory and they're constantly adjusting their actions to make it happen. So individuals infer each other's mental states by observing cues, which they can attribute to hidden causes. They use their own model of or models that they have learned to recognize about others. This process is known as generalized synchrony, entailing mutual predictability and can be interpreted as a collective minimization of free energy. In a group setting, individuals create cues in the world that direct other agents' attentions towards the same signals. Their models of the world align by sharing similar goals and similar environments and by reinforcing patterns of sampling reality.

So we could try to represent this through polynomials and morphisms. Each agent's interaction with the environment corresponds to a morphism and the collective interaction of all agents can be represented as a polynomial. So a morphism, if a polynomial p is defined, a morphism between two polynomials p and q consists of a pair of functions f and g such that f maps p of 1 to q of 1 and g maps p of i to q of the quantity i and p of 1 maps to q of 1.

We create a framework to further explore the cooperative and competitive dynamics among agents. For multiple agents indexed by i with boundaries p and shared environment boundary q , their interaction can represent can be represented by a morphism of polynomials which allows us to understand the shared potential in multi-agent systems and provides a foundation for further exploration of cooperative dynamics.

We allow each agent to predict not only its own behavior but also the behavior of its companions and the environment's response to their actions.

We can use shape-theoretic and topos-theoretic approaches to understand multi-agent systems.

A multi-agent system is a system where multiple individuals or entities interact.

So for example, a traffic system can be considered a multi-agent system where different vehicles or interact.

But topos is a category that has both spatial and logical structure allowing for the expression of logical propositions and deductions within it.

So we can use these approaches to construct a coherent understanding of the world from the perspective of multiple agents.

The internal universe of agents which can be seen as mathematical objects containing beliefs, perceptions, and predictions of individual agents are represented using topoi.

These internal universes provide a foundation for understanding and representing the complex interactions and shared understanding among multiple agents in this dynamic environment.

We try to construct a consensus topos by patching together the internal universes or topoi of multiple agents.

And the idea is to use the sheave-theoretic and topos-theoretic structures to build a shared understanding. So we can thus derive the synchronization of goal-oriented actions among agents over time and leverage

mathematical insights for category theory and generalized generative models.

This connection is the mechanism through which the group synchronizes over time and ultimately achieves their objective.

Agents within the system might possess varying levels of certainty in achieving the objective.

Some may choose to expend less energy temporarily, decelerating their progress towards the desired outcome.

This variation in certainty and approach leads to the emergence of different roles within a group, with some agents focusing on computational tasks while others monitor outcomes or engage in different actions themselves.

The nested and distributed structure of generative models in active inference allows for the integration of both short-term and long-term goals.

Agents with similar or compatible preferences can coordinate their actions by aligning their policies and working towards common narratives.

So this alignment reduces the overall expected free energy of their joint actions, increasing the likelihood of achieving shared goals and

facilitating the sharing of goals and expectations among group members.

They can learn, which allows them to learn a central role in shaping agents' preferences and strategies over time.

Agents can gather new experiences and update their generative model.

They refine their preferences and methods for achieving goals.

They can influence the formation and modification of shared narratives by having different perspectives over different parts of a goal.

Agents can also be a central role in a general life.

Here, the concept of general life synchrony is manifested when all agents have learned compatible or harmonized aspects of the overall generative model.

The synchronization is rooted in the retensional and potential dynamics of agents and can be more accurately described using mathematical models.

So in conclusion, we tried to shed light on the concept of shared potential goals and multi-agent systems.

We've emphasized the importance of understanding this concept as it plays a crucial role in the coordination and synchronization of agents' actions and intentions.

We could, in the future, continuously highlight the role of cues in the environment and how they can be simple like marks on the road or more complex like symbols and language.

The cues become carriers of passive synthesis, far pretensions, and far retention and become ways to minimize free energy as they minimize the amount of any individual has to compute.

It also allows us to delve into the concept of other evidencing in which the environment is in effect the self and others must learn to navigate in terms of others and self.

We've provided mathematical models to an equation to analyze dynamics of shared pretensions more accurately in terms of category theory.

We've explored the dynamics of shared retentions and pretensions in hierarchical generative models, which reflect the complex interplay of individuals and collective dynamics.

We've encapsulated the structure of time consciousness and how this allows agents to co-compute shared pretensions by mapping each other's cognitive pathways.

We've also developed a lot of ways to understand the context of a particular context and how this is.

So our approach basically begins an examination of Husserl temporal phenomenology integrated with the hierarchical active inference model, polynomial morphisms and polynomials.

This enables recursive cognition and prediction of not only an agent's actions, but also the environment's response to these actions.

So I think this comes to the end, Daniel.

I think I'm literally at 30 minutes.

Thank you.

Impeccable.

Thank you, Mao.

Yeah, great presentation.

Awesome topic.

So thank you for joining and looking forward to seeing how the work continues to develop.

All right.

Farewell, Mao.

See you.

See you.

This comes soon.

I think I'm going to be back.

See you.

Thank you.

All right.

Okay.

All right.

The next session is going to be the final session of this symposium.

It's been a really incredible journey.

And I think this final session is going to be great as well.

So we have a whole panel here and people can use their cameras, even though I'm not, if they want to.

Welcome everyone.

I'll just introduce the facilitator, Kurt Jai-Mungal.

So Kurt is a legend in the active ecosystem and has done incredible work with a variety of researchers and curious individuals in the space.

So we could think of no one better to host this conversation and bring it all together.

So Kurt, I'll pass it to you.

You'll keep an eye on the live chat.

Anyone can call it if they need any supports.

Otherwise, thanks everybody for joining.

And we don't see you yet, Kurt.

Yeah.

Would it be all right if I left and then came right back for whatever reason my camera is not working?

Sounds good.

Sounds good.

Okay.

I'll be right back in about 25 seconds.

Well, welcome everyone.

We will get started in just a second.

Hmm.

So odd.

I apologize for that.

Okay.

Well, you can hear my voice at least.

Yep.

Just give me a moment.

I apologize.

Boy.

Okay.

Well.

Well, we'll do what we can.

Okay.

So you all can't see me, but that's all right.

So we have this all-star panel here today, which to me means my invitation must have been a mistake, but I'll take it.

Thank you to the Active Inference Institute.

And I'll go around this virtual Zoom table and introduce everyone briefly.

Carl Fristen is the Wellcome Principal Research Fellow and Scientific Director at the Wellcome Trust Center for Neuroimaging and Professor of Neurology at University College London.

He's also the Chief Scientist at Versus.

Anna Lempke is Chief of the Stanford Addiction Medical, sorry, Medicine Dual Diagnosis Clinic at Stanford University.

Her popular books include Drug Dealer, MD, and Dopamine Nation.

Raphael Kaufman is CTO of Digital Gaia and is on the board of directors of the Active Inference Institute.

Bert DeVries is a professor at Eindhoven University of Technology, where he directs the BiasLab research team and also works with industry.

Guillaume Dumas is an associate professor in computational psychiatry of the Faculty of Medicine at the University of Montreal and the director of the Precision Psychiatry and Social Physiology Laboratory in the CHU Saint Justine Research Center.

He's also affiliated with MILA, which is Quebec's Artificial Intelligence Institute and Other Art, Science, and Consciousness Service Initiatives.

My name is Kurt Jemungel, and I use my background in mathematical physics to analyze various theories of everything that are proposed.

These include both the theoretical physics side of grand unification with gravity and dualities, other schemes, as well as attempting to understand what constitutes consciousness.

You can find the podcast by typing in theories of everything onto YouTube or whatever podcast catcher you have.

So my question to everyone, the initial question is, what have you been working on in the past few months, and what excites you about it?

We'll start with Bert.

You seem to be Smiley.

And your name sounds like mine.

And your name sounds like mine.

And your name sounds like mine.

I'm going to...

Okay.

So, all right.

So, my name is Bert and I lead a lab in electrical engineering department.

And so, our lab is called BIOSLAB, as you mentioned.

So, we are interested in Bayesian inference in general, but more specifically in doing it as fast as possible.

So, there has lots of applications outside active inference, also in signal processing and other control systems, but definitely, of course, also for active inference.

So, half our work over the past few months, but even over the past few years, has been on developing a toolbox to support or to get as far as we can go with trying to do real-time Bayesian inference or real-time free energy minimization and trying it in applications.

So, that's the work that I try to lead a team of PhD students who do the real work, of course.

I walk around with a cup of coffee, but that's what they've been working on.

Great.

And Professor Lemke, please.

Yeah.

Well, what I've been working on in the last, I don't know, year or so, is spending a lot of time thinking about how it is that a faith framework, in particular surrender to a higher power, improves people's lives.

And I've been really struggling to come up with a way of talking about this, that is inclusive, thinking about it, talking about it.

It gets to a kind of core piece of it that I'm interested in, which is not so much whether or not God exists and what the proof is or not of whether God exists.

you're muted yes that's going to be the title of my next book carl you're on mute i think that's a retro from the code possibly the future um i i generally work on what i'm told to work on so i'm just been trying to list uh what i've been working on um in the past few months interestingly um it actually starts with um daniel friedman and his um fascination with um along with axel constant for evolutionary explanations and speak to many of the scale free issues that we've been hearing about um in the prior session and so that's what i was working on really a scale free approach to understanding temporary nested processes as free energy minimizing processes as a kind of active inference and how that would lead to a variational synthesis of evolution and then i was told to work on belief sharing by mao and colleagues verses and worked on that in the context of synthetic language and tried to establish a really simple proof of principle that communication was an emergent property of any free ensemble free energy minimizing system um and the just open brackets just to make the observation it struck me time and time again uh during the session today how important communication is um you know whether we're talking about birds um passing messages or responding to uh you know you've got a new posterior or we're talking about the kind of communication um that man was talking about well everybody's been talking about it does seem to be a really uh and indeed the guy the guy um notion of over connectivity and too much communication or the wrong kind of communication um it does strike me that's quite central uh to a lot to the the best the theme of the um the workshop um um and um the specific uh presentations that we've heard um and then uh in the past few weeks working on stuff i think that burt would be um um more interested in which is um fast and frugal um message passing schemes that um enable the evaluation of the expected free energy not to suffer from uh the limitations imposed by deep tree searches so um possibly uh ashwin has spoken about this earlier on but um you're using ideas from dynamic programming backwards induction to try and um take the pressure off the the message passing to get fast and efficient uh evaluations of the expected free energy in sort of um sort of single unit uh sort of single agent uh active inference what have i been doing this week can't remember no that's what i've been doing great rafael kaufman please

just a moment

and can everyone else hear okay

what about now yeah great yeah thanks uh it's contagious i think uh no so i've also been drinking a lot of coffee as i've been uh caring for my now nine month old daughter so uh congratulations thank you so aside from sleep deprivation what i've been doing is um uh i i told you guys about this in the in the session earlier today but uh in a nutshell we've been working on what we call the the the guy protocol uh or the guy network which is this uh this um open approach for for building a uh common uh common engine you can think of it as a common runtime and uh in a common uh language for with the building blocks of of shared decision making um in the context of or the mad crisis and uh getting to get into a set of of building blocks for decision making that don't kill us all so that's that's kind of it

rafael how many how much coffee is a lot of coffee well it's espresso so i don't think it's quite comparable to it's like american coffee all right and guillaume dumas please yes sure can you hear me yes okay uh so in the context of action inference uh i have mostly focused on uh well the two big topic of my lab so uh precision psychiatry and social physiology so on the precision psychiatry side uh uh nilke boyton presented earlier in the symposium our work on trying to model theory of mind to being able to evaluate uh through digital phenotyping the level of sophistication of interaction in patients with neurodevelopmental disorders and that connect with also our uh interest in the social mind and how we can uh connect our understanding of how we deal with other people but with also ourselves and so that connect with other work at the mila with deep learning architecture with higher order function in humans where we're trying to combine different theories of consciousness such as global neural workspace and attention schema theory and in the context of attention schema theory there is this kind of recycling of self and other mechanisms so that's one uh big work and on the other side i'm very interested also in uh multi-agent uh systems and so with the natalie castle who's going to join us uh soon we've been working on creativity and the emergence of cultural norms in multi-agent systems uh with the idea of applying that to climate action and also a new project that is starting here with the uh nfrf of a big canadian project on indigenous knowledge and how we can think about new narrative around ai and typically a less solipsistic individualistic way of dealing with ai so that's the the two main uh field of research can you explain what you mean by less solid statistic less solid geez louise so it's not just artistic when it comes to the the californian view of ai because the narrative is mainly driven by uh us and california right now tends to circle around optimization profit i mean even open ai states that uh agi is about optimizing profitable human tasks they they put like profitable in their definitions but uh in the case of indigenous uh systems of knowledge we are more talking about community sustainability and the view already only in the application is less individualistic but also in the cognitive science point of view we we have like the the hardcore computationalism that would say

the brain is just like a computer in a very restrictive sense and so uh ai like typically the those transformers that are massively used right now uh are kind of like brain in a vat in a very silicon

uh sense while we are embodied systems that are uh constituted by our interaction with others and so

like in a way how we can think about artificial intelligence in this kind of more co-constitutive way and through a developmental and cultural lens mm-hmm okay now speaking of the brain as a computer we frequently frequently hear in these circles the brain as a predictive machine so where does this the quote-unquote brain as a predictive machine have its limits and also is that to be interpreted as implying conscious experience is a predictive machine as well okay and if not why not so i think carl you'd be a great person to start this off and then we'll go around the table again yeah i was mindful of mao's air presentation in the past half hour um and a notion of temporal thickness um and i'm sure that's got a lot to do with the necessary conditions for the kind of consciousness i'm guessing you're referring to you know that freedom from the moment um and if one reads prediction in its sort of psychological or protensive or pretension um nature in terms of being able to predict what will happen in the future i think that the um that's that that's going to be a sort of uh a key bright line between things that um do not possess a certain kind of um ventions and things that do um and that bright line just rests basically upon um well from perspective of active inference having a generative model that includes the consequences of your own actions in the future so just by having consequences you're now talking about um the future and the cause the consequences um in the future now become random variables therefore you have to infer them which leads you directly to the notion of planning as inference which means that bright line it's just the difference between things that plan and do not plan um and i would guess that that's where you probably want to start in terms of foregrounding the role of prediction um as being an aspect of self-organization and it's sort of reading under active inference that characterizes conscious things from uh non-conscious things namely the ability to plan does that make sense i have follow-up questions we'll get to them at some point but for now anna do you have do you have any comments on that and and bert as well afterward um well you know i'm i'm new to this field so i'm just familiarizing myself even with the language but i i can respond to the question the brain is a predictive machine in terms of the work that i do clinically in the kind of psychopathology that i see when i'm working with patients um you know we i work primarily with narratives the stories that they tell about their lives and one of the recurring themes i've seen through my work is that when patients tell stories in which they are perpetually the victims of other people's actions or the world they tend to then that tends to be a predictive model for them so that they will then go out into the world and unconsciously create scenarios which will perpetuate their victimhood and that part of getting into wellness is to stop seeing themselves as entirely the victim of other people or circumstance instead begin to appreciate what they contribute to their life problems so i guess when i think of the limits of the brain as a predictive machine i think in some ways one of the big limits is that it's a very powerful predictive machine that actually allows us to subsequently shape uh what actually happens to us and or our perceptions of what happens to us which then can perpetuate um a false narrative and i i'll just give one small example from my own life i've had um you know my conflicts with my mother and one of my beefs about her is that she's a very poor communicator and that whenever i email

her i either get a cryptic response or no response at all and it drives me crazy and about five years ago she sent an email asked me some questions i wrote her back and responded and it was clear to me that that required yet a response from her which i never got and that then perpetuated my narrative about her as a very poor communicator and many other negative things and then about three months after i sent that email i found the email in my draft inbox in my draft box so i had never actually responded to mother's email i hadn't actually sent the email and that was for me um just a you know a personal uh wonderful example of the ways in which uh we can our actions can actually um um be manipulated to uh support our models uh and perpetuate falsehoods about the world that we live in right and i have a quick question bert just for anna before we get to you so you use the word narrative there how are you defining the word narrative is it the same as model is it a sequence of events like what is the specific definition of narrative you know i never really thought about it in those terms but when i talk about narrative i'm talking about um the stories that people tell about their lives because that's sort of my data um and and also about models because what i've discovered about self-narrative is it's not just a way to organize the past it also becomes a road map for the future that's the language that i use but i see it maps very nicely onto url's a language of modeling the world i see professor de vries please um yeah it's it's it's clear that uh the ability to predict is is the essence of uh intelligent decision making um yeah i have a different background right i'm not a psychiatrist so i think about these things in different ways the thing that i think about when i think about prediction is i i would assume that it brain predicts far ahead that things get less accurate right if i predict that i want to go three quarters around around about i don't care about the centimeter where my car goes when i'm around there i just want to get in in the right lane and i would assume that the brain doesn't take much it takes less computations if you care about things less precisely but that's very hard in a computer we have this is what uh kind of kills us in uh in in i think in our current uh way of implementing active inference when we want to predict a deep half we don't care about the accuracy but we don't know how to do it much cheaper when things are less accurate and uh so that's still a that's a problem that we uh i think we're working on um it might be an uh it might be a key to building or to scaling of active inference agents if we can actually compute messages that we want to know that we that we don't care about if they are less precise that we also spend much less computation on them thank you now professor kaufman uh i'm not a professor but i'll answer anyway uh you just call me rafa uh yeah so uh there's uh so many really interesting avenues here and it's not often that i get the the the pleasure of discussing this kind of stuff with uh with this uh kind of diverse crowd but i do i mean i've been interested in uh in these questions uh for for a pretty long time and i i i think what comes to mind is like how how active inference as a lens uh for instance it enables us to to to get another sense of what non-dualist views on consciousness uh are saying when they when they uh get us experientially to to notice the difference between what we actually perceive and what between the different various different uh uh processes or things that are going on in our in our head and our tendency to like lump them all together into the same okay this is this is my this

is my experience that that i'm immersed in um so noticing being able to to to uh notice the difference between the the the the process of perceiving and act and and acting and what it's automated and the narrative or the various narratives that are going on uh uh in parallel in my head whether i'm aware of them or not and uh and our ability to to to just put kind of seemingly arbitrary manner levels on top of it to to try to make sense of it all into to force uh for static experience of having multiple narratives and multiple framings going on at the same time makes uh make sense of that experience you know in a um you know it that aligns with our presuppositions about about how things are so um i i think that's super interesting and also i also want to say like that uh like this this view of consciousness as being kind of uh defined by having a narrative or an internal model that's about self or that at least uh is about self and that leads as as carl said to planning as inference uh that's actually like super deflationary in a way that that people are not uh not used to thinking about it and uh and so it's an exciting it's opening the doors to all sorts of exciting interdisciplinary conversations to be had on the basis of at least i feel like better uh less talking past each other than we've ever had so i'm excited about that sure and how is it that you're using the word deflationary there carl i asked you about that when we had our podcast together but rafael is the similar question so you said deflationary view when we have narratives of self do you mean it leads us to something that consciousness is much more than sorry much yeah yeah i mean it's uh not necessarily much less than in the in the uh subjective sense but it's uh perhaps much much less than in the in the scientific sense so uh one example of like one not saying exactly that but but maybe something close to what uh daniel bennett calls heterophenomenology which is basically the the statement that the the sum of what one can say scientifically about consciousness is is um is equal to what can be can be uh what can be studied and and modeled and theorized and communicated and learned and uh agreed upon on the basis of objective data about

what people including you but there are also other people say uh about their experience which and of course like what can be measured up new correlates and whatever which is not the same as our not necessarily

the same as our experience of our directing what we think is our direct experience of being conscious conscious right so one way to look at it is that uh uh is that uh the science of consciousness doesn't necessarily have to put the primacy on on people the first person people think they have qualia uh that that we thought in that sense this deflationary right it's um yeah you have to explain why people believe they have quality not why people have quality because it's not it's not necessarily a scientific truth that people have quality right professor dumas uh yeah so well on the limits of brain as predictive machine well which we should be always careful to not move from uh one uh map territory fallacy to

another one for sure so we need to be uh skeptics and avoid to refine the methods and well the this world symposium show how this metaphor is super useful and and fruitful but um i can see like two main uh limits or at least uh things where we should be careful so uh following what uh professor lemke said like in psychiatry i think the the looping effects and the way people picture themselves uh can be a very

unpredictable and so the the way uh we think about the brain as a predictive machine i think prediction for patients would have different signification of what we mean by predictive machines in the context of active inference and in general also like uh the i i can see that certain uh psychologists or anthropologists would have a big appeal all of a sudden for uh active inference and pre-energy principles but taking the the words uh as uh directly what they think the word mean well it needs to be we need to be careful in the way to communicate it and typically uh following also uh what professor devris said about uh predicts prediction as being super important for decision making i think like uh uh we should be careful about the the weirdness of uh cognitive science and how uh it doesn't necessarily expand to uh non-western educated the industrialized country where maybe the cultural value is not about optimization of your decision making or your uh profit and so that's the first thing where i think we should be careful it's more like a matter of how to communicate the theory in that case not necessarily a limit of the theory itself but the second one the third one is uh about the the equivalence or not with other frameworks so like uh uh i can see how we had the mao uh albacin talking about uh category theory and i'm very curious about right now among the different frameworks that are out there how can we define what is equivalent or not with active inference uh to be able to to see those limits uh maybe we are leaving what happened with quantum mechanics in the in the early uh 20th century with different interpretation with different tools to model quantum mechanics and here with artificial intelligence and cognitive neuroscience we have also all those frameworks and uh i think uh the limitation would be then to not take this interpretation as the only one but uh uh try to have a crosstalk and a differential uh diagnosis of which one is the best for explaining what okay now my role is is the moderator but i would like you all to to speak to one another so i'm going to ask a question that will serve as

well you each will speak on it but as the other person is speaking as sorry as the other people are speaking just think okay is there a question that i have or is there a comment that i have so the question is what are some of the recent advancements or breakthroughs in your respective fields that you find particularly

promising and well recent let's say 2020 till now so we'll start off with rafael you're smiling and you look like the champion oh i i was thinking what is my field right because uh we're very much in the in the um in the last mile of applying the the wonderful stuff that y'all come up with and uh and uh making it uh useful on the ground but so i'm i'm extraordinarily excited about research like what uh what bird is doing and uh uh uh i feel like as uh as guillaume was saying we're also starting to see some uh some convergence on on different um different ways to get to to the same uh at least the same shape of answers and in some cases even to the same uh the same uh results so one example of this is uh work that uh um chris chris fields carl and others have been doing on uh on the the quantum uh uh basically explaining quantum information theory uh and uh and how it relates to to to the free energy principle looking at the free principle as the as the classical limit of of quantum information theory and i don't pretend to to understand all of it but as somebody who has who's coming from a quantum physics background uh just uh uh being alive to see this this kind of conversions happening uh which as i mean as as uh as guillaume said we're not we've been at this at this business of like post-classical what's what the hell's going

on for over a century now and it's nice to to be at a time where again we're stuck we're starting to talk less about each other and that's that's from a theoretical perspective from a practical perspective i think just uh just finding that we have all the building blocks to to uh create uh fast interpretable uh um um reliable and aligned um decision making systems which includes uh ai systems autonomous systems that have all those characteristics and that it turns out that all the things that the reinforcement learning and uh and deep network networks people talk are very hard or impossible or sort of in the realm of of what we can do uh looking at things from an active inference lens and uh vice versa a lot of things that that that uh active inference models have incorporated so far there it turns out if you if you toss in a neural network approximator here um and um you know some borrow some other you know massively parallel computation techniques uh it also becomes feasible so it's it's converging yeah for me it's what's remarkable is that there's so many domains of of what we thought was exclusively human or would be exclusively human for decades that in just the past couple years robots or computers seem to be just as good if not exceeding us and i don't know if that's promising or worrisome but anyhow carl please if you don't mind answering the question and then we'll go anna then bert and then dumaz well actually just reflecting upon your um one of your observations i think it's very difficult to identify one thing uh you know in a sense um what is impressive is the diversity of advances and applications um and i just say that because that's what i was thinking you know over the past six hours just listening to your amazing presentation after a presentation and just you know noting how diverse you know and yet there's this common thread this common commitment uh basically saving our curiosity and and using the the tools that naturalize that kind of sense making curious behavior and communication um that inherit from sort of either maths or category theory or as rough notes now quantum information theory but just to pick up on a couple of things which are relevant to this conversation so the work with chris fields um yeah it sounds um lovely and exciting to bring quantum mechanics into um active inference but that's not the move that i think chris is really wanting to champion the move i think is something that we've all been addressing in one way or another which is um leveraging the scale free aspects of these this principled uh approach to self-organization and hopefully self-organization to some kind of generalized synchrony um so that that's where the quantum information theory gets into the game it is scale free and indeed he will go further and say it's completely background free and everything is constructed it's just uh uh so i think that that's a lovely move um because once once you've gone scale free you then start to ask deep questions about how you couple one scale to another scale and in a sense ecosystems is just that how do the you know the denizens of an ecosystem um how is it constitute how is it co-constructed what is the structure of it these are all questions about how one scale links to another scale so i think there have been lots of advances in that direction in many many different fronts um and you can read that either in terms of coupling different sort of spatial or um yeah spatial scales um but probably more importantly sort of temporal scales

and you see that you know wherever you look um um you'll just come back to um what do you mean by a narrative um it is exactly i think as i said it's just a plan it's just a story but notice the story has a temporal aspect to it you know i have narratives about being a good person being a good father being a good scientist i also have narratives about you know i want my cup of coffee or i have to go look after that if i don't but ralph has to go look after the child um so yeah we've all got narratives at very very different time scales and of course if you're um yeah i just came back to bert's example of you know i'm an autonomous vehicle and i'm sentient and we're five years into the future and i have to drive around the roundabout you know what temporal scale and what kind of temporal course graining to define the narratives necessary with narratives would be appropriate for that kind of situation um and um the you know the the ensuing planning so um so to my so a short answer to your question i think there have been many many advances um they have um i think what they've had in common is basically transcending either different domains but in particular different scales of application i also agree um with the notion well another i think important and pragmatic advance um is something that bert mentioned which is democratization of this technology i think ralph also hinted out you know this is the time you start using this for the for the for the common good um um so i think you know the things like um rx infer and pi mdp and uh and i didn't know about the the gaia project uh but it sounds as though there's been great advances there as well so this kind of democratization i think is really important this sort of socialization where everybody can play uh and start to sort of um not talk past each other i think that that's a that's a very important uh a very important advance and did you say the word scale and variance or scale independence i said scale invariant um i actually said scale free no i shouldn't have done that um i meant um i i said scale uh scale free so the the idea that you can apply exactly the same mechanics and literally for example say from bert's perspective the same kind of message passing at different space time scales or um at different levels in a hierarchical model so i've actually got a question for bert um in terms of reactive message passing because reactive means that you don't have to um prescribe the scheduling but you in in addressing the problems or the issues that are entailed by um having to specify the scheduling of talking or of message passing um you you're um you're bound to deal with time and in a scale invariant context or nested with nested scales for example you have to deal with the term uh the separation of temporal scales so i i think there's going to be a very important generic question which technically bert will have been thinking about furiously for the past few years but i think implicitly we're all going to have to be addressing soon which is the um how do you put that the timing of your messages when you make a move or when you listen to a patient or um um when you actually pass a message in on a factor graph um how does how are you going to how are we going to be able to put the separation of time scales into the architecture in a way that speaks to this uh the scale invariance you know the gear project for example how do you integrate um live feed from traffic flow um um sensors with uh fluctuations in the climate you know these kinds of this kind of data comes at very very different uh temporal scales and yet has to be assimilated and modeled um you know in a way that is also i think has to pay pay you know due

courtesy to that separation of temporal scales or could you please recapitulate the question for the audience and then begin to answer it okay um yeah the issue is if you have if i mean active instance agents are nested agents and the higher levels supposed to operate on a larger temporal scale but they should they're also working at a lower resolution right if you if you look far ahead you don't care as much about precision not at the centimeter level when i go around around about i don't care the centimeter where i land but if i'm but for the next few milliseconds i do care about because it may mean the difference between getting in the ditch or not so um i don't want to send so the higher level i want to look very deep ahead but i don't want to send every millisecond a message to go to to look let's say a minute ahead because so i need to space it out a lot but then i may miss things um so preferably you would just send inaccurate messages but that only works if you actually have a method to also let's say use less uh computational power to compute a less accurate message and we're not good at that yet um there i think i mean what we are thinking about is there's a new field a new field but there's a field called probabilistic numerics where um we are used in in math to just compute everything very precisely or as precise as possible and and and do not care about how much computation we spend on it um so in probabilistic numerics um i i hope we can leverage this for uh for message computations um i would like to spend let's say proportionally less computational power on uh on the accuracy of a message one way possibly would be to consider a message a latent variable that has an in uncertainty by itself um but i i don't have a a totally clear answer for for carl because we haven't solved that either um but there there is a problem in let's say what we do on our i mean our computers weren't so completely different from computers that from let's say from the um from the from the brain that uh yeah um we're spending we're spending just too many computations on messages that in the end are very very very inaccurate and that's a problem in uh in what we do on our computers currently um yeah so i don't know how we go i mean what you want to do with this on the higher level if you you want to look maybe um um 10 times farther ahead and spend about the same amount of computation on the

lower level that's sort of the uh our goal and uh but but we don't have an answer for that either at the moment and and anna so please feel free to comment on or ask a question to anyone yeah well i mean i you know i don't i don't have uh anything to contribute unfortunately to how computers work but i can tell you that this idea of temporal scales is something that we face often in our work with patients um for example addicted patients are very focused on short-term rewards and it's it's a fact in fact their ability to control how they feel in the moment that is partially what drives the addiction so when when i try to adopt your all's language of minimizing surprise or minimizing free entropy that's one of the the things that people are trying to do on a short-term time scale when they become addicted so i have a young woman who is addicted to nitrous oxide which has a very fast onset of intoxication over the order of seconds and a very fast offset and she says that that's exactly what she likes about it because she's controlling it second by second so when we work with patients to get them out of that short temporal horizon we actually rely more and more on action and having them change something in their lives namely abstain from their drug of choice for long enough to kind of completely reset uh their brains and to allow

them to see this longer temporal because when they're chasing this short control they're actually not able to see themselves in the longer narrative arc of their lives so that's that's what comes to mind for me i don't

think it's going to be helpful to people who are trying to build computers but that's that's the kinds of interventions that i make with humans i have a question about that so you mentioned control in the short term and you've also

mentioned that you study the positive effects of having a higher power in your life or or surrender

okay i kind of gave the punchline away by saying surrendering to the higher power what where i was going was okay what's the association between seeing yourself in the largest time frame and a higher power and then also is there something that is akin to giving up control when you yeah look farther and farther and farther into the future that's a great yeah that's really at the heart of what i'm very interested in because it's a real paradox right it's this kind of locus of control within ourselves that really in modern culture we think is a great thing but when that's taken to an extreme and one example of that is addictive behaviors it's very bad for people and for communities and so what can be what can pull people out of that is this kind of surrender to a to a higher power giving up that locus of control locating that locus of control outside themselves not not necessarily like in a theistic sense um one of the things you know that they talk about in alcoholics anonymous for example is um you know you don't have to believe in god it just doesn't it's just not you you're not you know you're not driving it and so i would be very curious from the perspective of your all's understanding of active inference and the free energy principle and how the brain works why is it that um sort of embracing our inability to control what happens in our lives can actually be the very source of healing especially embedded in this really kind of over controlled yeah i would even go so far as to say endemically narcissistic culture like i'm really curious i don't want to take the conversation you know in a direction please please that's a fantastic question so if anyone has a comment on that please i have some thoughts uh and i think this applies um supplies both at the personal level and at the at the global level and i think it has to do with the the my quote from earlier uh brought by edward fullbrook that uh if you're falling from a plane yes maybe an altimeter and some instruments uh might be useful but what you really need is a parachute right so we have this um uh we have this uh presupposition that that are whatever like framing we have we have operating in our day-to-day is going to be okay this this is the right framing and it's gonna it tends to be the this uh this rational scientific uh framing of okay very very linear uh cause and consequence for for for most stuff and it turns out that even like if you inspect our our day-to-day behavior you know bigger more complicated models are not necessarily better which is where we get uh the success of heuristics under a bounded computation bounded rationality uh and uh if you scale it up to to uh eight billion humans interacted in in uh in uh resource-concreened planet you know world of possibilities but also of of challenges then you just you have to we have to like drastically lower our our uh our bar or yes how much before we have but also even like how much what was where does uh information gathering reach diminishing returns where does modeling reach uh diminishing returns uh there's a whole literature on uh in the business

world about the expected value of information how much you should actually invest also in science is also uh known as uh optimal experiment design we're basically you know acknowledging that uh that you have a limited budget in terms of how much you can act and how much you can probe and how much you can

how much time you can spend thinking about stuff and uh i think what we're doing when we feel like burned out or or uh exhausted from from overthinking uh we're innately feeling that that okay we've gone too high and we need uh we need to to uh give ourselves a vacation give ourselves some uh some uh some free time here i think that doing this on a principle in a principled way uh we're just not not just like taking the heuristics and the signals that uh that we inherit from uh from evolution but actually being able to figure out uh collaboratively and uh and with with some rigor okay this is uh we don't need to know exactly how much how many degrees the world is going to warm uh by 22100 uh in order to know that maybe it's a good idea to to start doing something about the amount of of carbon in the atmosphere or whatever right uh and so i think this leads to this this this no knowledge this idea of uh of a real knowledge economy and of things like abstraction as a service how can you actually um how can you actually build in um building this this kind of like sophisticated and uh and um sophisticated uh translation layers that take some of this burden from from ourselves as individuals and even as organizations right uh and just put it out in the world as as value-added services which is what they are um um to make this is um to make this is um to make this is um to make this is um to make this is um um to make this is um to make this is um to make this is um to make this is um progress that were very interesting. We heard already about the formalism maturity at the mathematical level between multi-scale and scale-free aspects. I really liked also the development of a more multi-agent perspective of active interns. It's very interesting, like especially like for instance, the seeing multi-agent systems we just heard like even society as well, like as many as one system or many as many subsystems and how we can use maybe those formalisms to also deal with policymaking. That's very interesting venue and the emergence of norms, culture and ideas or so because one thing that I'm still struggling with in the case of active inference models is like how to get outside the checkerboard. You can put a lot in the model a priori and how you make the model creating new stuff that is not backed inside at the first and on that like one very interesting breakthrough was the application of active inference to morphogenesis. I really liked the work that has been done on that and on the technical aspects, I think the two main focus that I love are all the deep active inference and how to scale up active inference because it's a strong limitation for the adoption of the formalism if it's not scaled enough compared to other framework like deep learning. And then the link with empirical data. I really like for instance, the work of Ryan Smith and how to connect with the extra physiology and clinical data. And I think it's also something very important to anchor the theory in the real world and have like a falsifiability and empirical validation of those models.

And what are some of the applications of active inference to morphogenesis?

So, well, I'm not the expert here, Carl would be the best to answer that. But I was referring to the work with Michael Levin and how a model can help to create a sort of embryogenesis.

So maybe Carl, you can explain better than me.

Please.

Yeah, I don't have a reputation for explaining things very clearly.

But yeah, that's absolutely right. And it was just work with Michael Levin and colleagues showing that you can get quite expressive and biomimetic pattern formation and movement of different cells into an organization,

often described in terms of morphogenesis, simply by communicating your beliefs.

So I'm coming back to this sort of cross cutting theme of communication. So if you just broadcast your beliefs and you're a little cell and you have,

and there are a little ensemble of cells and they all have the shared generative model that includes,

if I was in this position, I would sense that. Now they all have exactly the same generative model, the same predictions, the same expectations.

And they're all broadcasting their beliefs about where they are.

And the free energy minimising solution is just when they're all in a place that they, such that they receive signals that they would expect to receive when they're in this place.

And of course, if that is the same for everybody, there's only one arrangement where each cell finds its place.

So basically it's just knowing your place is an emergent property of making the world mutually predictable through communication.

So it's a deflationary account of morphogenesis.

But the game, I thought, you know, one of your units is precision psychiatry.

So I thought you're going to talk about precision.

So I'm going to do a game now.

He talked about morphogenesis.

I'm going to talk about precision now.

I think that's a really nice way just to pick up on themes, which everybody's just mentioned.

In particular, the pathology of precision.

And by precision, you can read precision in the sense that Bert was talking about in terms of, do you use an unsigned integer or a double?

You know, how much can I coarse-grained my numeric representation?

Or you can use it in terms of coarse-graining in the sense of the renormalization group.

You know, it's just chunking things in hours or years as opposed to milliseconds and minutes.

Or you can look at it in terms of a statistician describing the reliability or the inverse variance of a signal.

And of course, we have to estimate that.

When I say we, I mean, your agents and statisticians and act accordingly.

And certainly in the active, in the work of Ryan Smith on addiction, much of the sort of mathematical explanation for these addictive locked in OCD like phenomena rests upon a failure to get that coarse-graining, that precision estimation right.

The reason I wonder whether it would be just worthwhile revisiting addiction and psychopathology or certainly pathology of behavior from the point of view of getting precision wrong.

And certainly assigning too much precision to low level processing, which is what Bert wants to avoid doing.

It just strikes me that that kind of story also has mileage in terms of why we are in a state of paralysis when it comes to climate change.

Because I also noticed, Guillaume, that you talked about climate action.

I've never heard that before, but that seems to me to be the important thing.

Why isn't there any climate action?

And Anna, you know, it would be like you going to the clinic and finding some of the Parkinson's disease.

Why isn't this person moving?

You know, and interestingly, the computational explanation for Parkinson's disease is assigning too much precision to the evidence that you're not moving before you move.

So, you know, if you don't, if you can't ignore the fact that nothing is changing or your prior beliefs, your predictions that I'm going to stand up or going to initiate walking.

Don't get a looking because they are immediately canceled because you've assigned too much precision to the lower level processing.

And I'm just wondering whether that kind of pathology is exactly what the, you know, what Ralph is trying to reverse by having, you know, models at hand, recommendations at hand that provide a more coarse-grained view of things, a deeper view.

Anna, would you like to comment on that?

Yeah.

Well, I just, I love, I love this idea of the pathology of precision because I think it, it manifests in so many different ways.

You know, not just among my, my patient population, but I think it's almost like a, like a cultural sickness in a way.

You know, the ways in which we, we seem like obsessed with certain types of data and we're missing the big picture.

So I'm going to, I'm going to think more about that.

I'm going to read more about it.

I really appreciate the discussion.

It's interesting for me.

And Anna, when you're referring to the pathology of precision, do you mean so in a more conscious sense that we're over-evaluating something that we don't need to?

Whereas Carl, you're referring to it in an unconscious sense, like the brain is putting too much precision on something because in Parkinson's it's, it's not like you consciously are putting precision in a place.

Well, I think you could look at it sort of like in both cases.

When I think about addiction, you know, people aren't doing it consciously.

It's that they really do see this as adaptive and healthy.

And also even on some level they can't do otherwise.

And they're not able to see the true impact of their drug use on their lives.

They're genuinely not able to see the negative consequences.

I mean, that's, that's what contributes to getting caught in that vortex.

But I mean, you could also see it as part of what's happened culturally, like, for example, like the whole wellness industrial complex, the way that we now count ourselves, you know, through all these different devices.

And if we could just count our breathing and count our heart rate and take supplements, you know, then we would somehow reach some levitating state of precise wellness.

And I don't know.

I mean, you know, just kind of, this is all, this is all kind of new, new ideas for me.

That's interesting.

So you believe that we can be inundated with health data and that that's detrimental to us.

Oh, absolutely.

I see that all the time.

Uh huh.

So for me, I used to have, okay, maybe I'll take this out of the, well, this is going out live.

Okay.

Let me be, let me figure out how to say this diplomatic diplomatically.

I used to have a device that would measure my heart rate.

Let's say that.

And, and my sleep, and instead of improving my sleep, it led to me becoming obsessed with it.

And then noticing, oh, I didn't sleep.

Well, I must not be feeling good today as well, because apparently there's a high connection between how your sleep quality.

That's exactly it.

Plus added to that, the other layer that I should be able to control it.

Right?

So with all of this data and information, I should be able to, I don't know, reduce free entropy or whatever, how it's produced surprise as you guys talk about.

Mm-hmm.

Mm-hmm.

You know, and I think that's obviously, that only goes so far, and then can actually contribute to our misery because we're, why aren't we all, you know, levitating like the Buddha or whatever?

You know, when we have all these tools and we can pay attention to all this data.

So.

I see this also with people who have productivity tools.

So, and not only that, but mechanical keyboards, let's say.

Some, the reason why I don't have a mechanical keyboard, even though I think I'd love it, is because I know that, why the heck do I care about the clicking sound of a keyboard?

But if I got one, then I'd be like, well, what's the difference between clicking sound A versus clicking sound B

versus clicking sound C?

And I'd become obsessed with the trappings of productivity that is sharpening, so-called sharpening the axe, rather than cutting down the tree.

There's this phrase that is apocryphal and it's attributed to, I think, Lincoln, which is that he'd spend 80% of his time or half the time sharpening the knife than cutting the tree.

And when you just, if you just think, and then this is just echoed in productivity circles, but it just, it can't be the case.

Why would you spend so much time sharpening your axe?

Like, look at anyone who cuts down a tree with an axe.

Most of the time, they're not doing that.

Anyhow.

So if anyone has any comments on what was just said, please.

Can I just talk about what I find exciting in my field about accident?

Sure, sure.

Yes.

I read, it was a long time ago, but there was a paper and it's about that it says, well, active inference is not really a scientific loop because it's biased.

And, and I read that and the, the, the, the, the, the, the sound of the paper was kind of, so it's not good.

But I think, you know, active inference is, is, is, is maybe not a science loop.

It's an engineering loop because there's bias and we need bias in engineering.

We need to make stuff.

We need to build stuff.

We need to have a bias.

So it's an engineering design cycle.

I see that everywhere around me.

Um, and active inference could be, um, a, a, a complete breakthrough in, uh, in engineering, right?

Um, the fields around me are signal processing.

I am myself in a signal processing, uh, department or group and, uh, everybody builds algorithms.

For us, I mean, in active inference agents, it's inference over states.

Then the floor below me, they build control systems.

Well, it's inference over actions.

Then other people are working on machine learning.

It's inference over parameters.

Active inference could be, and you'll like this, uh, Kurt, it's a very deflationary, uh,

uh, view on, uh, on engineering because everything is just inference.

And so rather than building algorithms everywhere, if we become really good at implementing energy minimization, we will, we will be able to build, uh, a great engineering design cycles.

And, and we'll be engineering better machines for, for, for, you know, for medical procedures

or for other things that are important.

So it has a tremendous, uh, application potential in engineering.

Um, in, in engineering, I think in most, in many fields, people have sort of drifted in different directions.

Control theorists have, I mean, they do almost the same thing as signal processing people, but they speak a different language now.

And, you know, so, uh, and everybody, and, and, and signal processing people, it's a completely different group from the machine learning people, but it's all information processing.

And this brings it, uh, this field can bring it together.

Uh, so I think it's, uh, if, but the thing is that in order to, to, to make it successful in engineering, we need to build an application that, uh, well, that impresses, right?

That it's not like a tic-tac-toe thing that it really needs to impress people.

It needs to be better than, you know, that some other control systems.

Um, but once we do that, I think there is tremendous, uh, application potential because there haven't been enormous breakthroughs in signal processing and control.

The last big breakthrough, I think was Kalman filtering and this was 1960s.

And, and I mean, it's, it's kind of, uh, it's, it's kind of funny that the essence of what we do in active influence is also Kalman filtering.

So, um, that's, uh, I think there's tremendous, uh, opportunities for, for what we do.

Here for, uh, for engineering.

So that's why it's, that's why it's exciting to me.

Adam, do you mind expanding on what you said about acting and therapists should be helping their patients with that?

Oh, uh, sure.

Just so, I mean, one of my critiques of, uh, you know, mental health treatment today is that there's not enough, um, there's not enough encouragement of patients to actually go and act differently in the world as a way of gathering data.

Instead, it often ends up being kind of this world building, um, between therapist and patients, not, not necessarily ultimately adaptive in the world.

So I was really just kind of responding to what Raph was saying that we need to act in the world.

I think that's more true now, um, in, you know, modern rich nations than ever before, because we are so incredibly sedentary and interacting.

Of course, we're interacting with a virtual world and that's, you know, good and bad, but I mean, I think we, we need to be actually acting in the world.

Hmm.

And so there's, there's different forms of therapy as you, as you know, there's talk therapy and then there's also cognitive behavioral therapy or psychotherapy instead of talk therapy, but, but cognitive behavioral therapy, as far as I understand, focuses on, on the actions.

And, um, I mean, for, you know, again, treating addiction, like you're, you're not going to really get that far with cognitive behavioral therapy or anything that's focused on just emotions and cognitions. People have to go out and actually, um, try stop using their, stopping their substance or their addictive behaviors and gather data from that experience and then come back and process it.

So Anna, in your field, and this question will go to everyone, but in, in your field and what you study, where is the largest gap that you would like to see closed?

Um, now we have more and more young people coming in with depression, anxiety, suicidality, addictions of all sorts.

These are people who have really privileged lives in many instances.

You know, what, what is going on for people?

They're not having experiences in the world.

So.

I think, you know, the, the, the co-created sort of models through healthier communication that allow people to feel part of a community and also to have like truthful co-created narratives, trying to use the language here.

[illegible]

Is it affecting the population the same?

Or is it affecting people who deal with abstractions more and more?

And then there's some people whose work it is to do something physical, like running or swim.

Or are you noticing that there's some broad trend?

Well, the broad trends that are out there are just correlational.

But the more time that people spend in the virtual world, the more likely they are to suffer from depression, anxiety, and other mental health problems.

You know, people haven't really been able to narrow that down to specific content, but they have been able to save just the sheer amount of time that you're spending online increases your risk for certain poor mental health outcomes.

Okay.

Okay.

Now, Carl, if anyone has any comments or questions, please just raise your physical hand.

I can see that.

Okay.

Raphael, sorry.

I was just going to say that I think another notable trend is, and I just saw somebody say this on YouTube just yesterday, that young people are disproportionately affected by things like climate grief, because they're the ones that are going to be alive to deal with it.

And I think that applies more generally that I know Peter Sandy already like 30 years or something ago, wrote about the inescapable network of neutrality reality that what we do affects each other.

Right. And we built, we took advantage of this huge resource buffer that's called the biosphere and earth to pretend that it didn't for quite a long time.

And, you know, got, got a lot of mileage out of it, but now we're at a point where we know that there's, there's a whole generation of people that are coming to grips with the fact that I'm going to stop myself from seeing a square word, but, oh my God, we need to, we actually need to change everything about everything, about everything that we do.

And, and we need to do, and, and we need to do it fast.

And, and by the way, that it's not just what we do in, it's not just what we do out there.

It's also what we do inside, how to get ready for, for how to show up for life internally.

Right. So no, no wonder that, that it has that impact.

So my, myself, I've dealt with anxiety and, and a lot of, of, of other things we've had conversations about what, what are we doing bringing a daughter into this world?

And all these kinds of things.

And I think it's, it's only natural that, that it's, it's coming to a head in this way right now.

You know, we have your hand up and I can't see you.

Yes.

Yeah.

It's connect with what has just been said.

So, and your initial question about the, the gaps that needs to be closed.

I think like a, a scale three model of health and mental health particularly would be great.

Like we are in silos in biomedical research.

And the fact that someone is having depression can come from interacting genes as much as interacting people.

And, uh, also, uh, is related to climate change and so on.

So how we can have a new health systems that doesn't deal with those silos and integrate those different scales to me.

It's like really, uh, a big challenge, but, uh, a change that, uh, current models and work that we can see, uh, go in that direction.

And I'm very interested to talk about that.

Bert.

Um, yeah, the, um, well, I, I, in, in my field in engineering, active inference is not understood, um, because almost all papers, uh, written by, by neuroscientists and they're hard to reach.

So I was really, uh, happy to hear today that, and I think it's Sanjeev, uh, who, who is writing, uh, an engineering book on active inference.

So that will, I think that will really help, um, that together with the availability of good toolboxes for implementing active inference should make a lot of engineers, uh, much more enthusiastic about, uh, active inference.

Because that's, uh, it's not something that is, uh, uh, it's not understood at the moment in, uh, in engineering circles.

So I hope that, uh, yeah, the book will be good, uh, because I'm enthusiastic about that.

And Carl, where, where are some gaps in the, in your research that you'd like to see addressed?

Um, oh, there are gaps, there's a whole empty space out there yet to be, um, explored.

Um, but in terms of, um, in, in terms of what seems to be emerging from, um, the session and specifically the past few, um, answers.

Um, it, uh, it does seem to be, um, important to have, um, this very generic, just to take Bert's, um, sort of line that this is just one deflationary simple and probably the right way to understand stuff and to make recommendations or to, um, describe people's actions possibly to themselves in a therapeutic, uh, context.

Um, and as such, it should be push button technology, um, and it should be democratized and socialized.

And, um, I, I think that's the challenge practically.

Um, and one, one may ask, why would you want to do that?

Um, for me, there, there are two, um, clear imperatives.

Um, one is, um, very abstract and it's not really my, um, within my comfort zone.

And the other one is in my comfort zone, but the one that's outside my comfort zone is, um, is, um, this notion of, um, interactivity and hyper-connectivity in the, and, and the meta crisis that we, we heard about.

Um, um, um, um, and, um, uh, Guim also referred to this in terms of what he was trying, he was trying to distinguish between a, a Californian notion of optimality and another kind, another way forward.

Um, and to me, it's, it's, it's, it's, you know, a stark contrast with growth is good versus sustainability.

Uh, and of course the maths of the free energy principle is just about sustainability.

It's just the description of the physics of systems, random dynamical systems that self-organized to some non-equilibrium steady state.

And that is what we are.

Um, so, you know, for me, there's something deeply, uh, if you like, apt about the free energy principle and its chronology, such as active inference in application to ecosystems and lived ecosystems and realized

ecosystems.

Um, so if that, if those basic principles can be brought back, um, into globalization, uh, into the, you know, into the market, into fintech, into, um, social media, into politics, into your climate action.

I think that would be a good thing.

Um, I'm just mindful of, you know, this struck me in a number of the, um, the presentations today.

Um, if you remember before Bert was saying, if you look at the brain, which is a really lovely example.

A really lovely example of a self-organizing system to a non-equilibrium steady state.

Um, then it's empty.

And what did he mean by that?

Well, it didn't mean you're empty headed.

What he meant was it's incredibly sparsely connected.

Now that tells you immediately that a pathology of connectivity is hyper-connective to over-connectivity, which immediately, um, well, um, it made me very alert to, um, the presentation of the meta crisis.

That one of the first three things that was, uh, uh, underwrote them, the, the, the meta crisis or, you know, the current crisis where we're, uh, content tending with is a destruction of that sparse, delicate connectivity that defines thingness and defines ensembles of things technically in terms of Markov blankets.

So if we want a world in which lots of different things can coexist in some kind of generalized synchrony in a sustainable way, you need sparse connectivity.

And the pathology, the thing that will destroy that is over-connectivity.

Um, so it seems to be very important that we get that into, uh, into play in terms of, uh, machine learning, um, artificial intelligence, politics, FinTech climate change.

And the only way it's going to get those epistemically by, uh, equipping people to actually, um, build their own little models and ask their own questions.

You can't tell people this, they're going to learn and they're going to learn it for themselves.

Uh, just very quickly, except I'm sure we, yeah, we've only got a couple of minutes left.

Um, uh, the, the, um, the other agenda, which I'm more familiar with is exactly Anna's and, and Guillaume's agenda, which is, uh, making this work in the context of, uh, um, of neurology and psychiatry.

Um, so if you can democratize, um, and socialize this way of describing things so that people can now build models of that particular patient in the other use of precision psychiatry, um, I suspect that Guillaume's, uh, unit was, uh, called after.

Um, so to personalize medicine that is really personalized in the sense that it, you know, you actually have your digital twin of the, of your behavior.

And then you've got that, you know, you optimize your digital twin to become a model of your patient.

And then you can start to do experiments on that, on that model, behavioral interventions, or even share that model very much in the spirit of CBT with the patient and say, look, this is you.

This is what would happen if you went out and did this.

And this is what would happen if you went out and did that.

Um, you know, that, that to, to my mind, and indeed that was the initial motivation for much of this.

Work was actually to build, um, observation models of psychiatric conditions to work out both the, um, the, the, the pharmacological and physiological basis.

And the disruption of the overly, the, the pathology of precision and message passing on the factographs that are our brain, even though they are very empty on the one hand, but also get that behavior.

That, that, that, that, that key thing that Anna was talking about, that active engagement with the lived world, uh, into the, into that model.

And hence active inference.

And just to conclude, you know, um, that, that, that, that activity, that sort of physical engagement, that embodiment, that sort of, um, for ease and everything else.

Um, I think it's really coming to a head now in terms of people's, um, you know, after the large language model, after the chat GPT moment, people, the bounce back, uh, has been what's missing.

You know, what's not there.

And of course, what is not there is agency and embodied engagement with the world.

Um, and that's why I think there's still, there's still a lot of work to be done, you know, in bringing artificial intelligence read as active inference to, in a way that matters to people who, you know, you know, the people who can make a difference, which is basically everybody, but specifically politicians and doctors and engineers and the like.

Hmm.

So you all now have 30 seconds to 60 seconds to speak directly to the audience.

What is the message that you, what closing message do you have?

You're speaking directly to someone who's listening.

They're a curious person.

They're interested in active inference.

They also want to lead better lives, hopefully, and, and, and do something propitious.

So what message do you have for them?

Anna, we'll start with you.

Yeah.

Um, I'm just going to say what pops into my mind right now is that one of the things I have learned from my patients who are trying to get into recovery from severe addictions is something that they call the set aside prayer.

Where they set aside all of the notions that they have about how the world works and try to be completely open and receptive to information coming into their minds.

And I think that's just a wonderful, um, a wonderful frame or concept for all of us, uh, living in the world, um, to periodically just take a moment and take these models and just say, you know, everything I think I know about the world, I'm going to temporarily suspend it.

And I'm just going to be open.

And when that happens, we can be present and learn in a way that it's not possible when we're just trying to reinforce our models.

Great.

Fantastic.

And Bert, then we'll go Carl and then Raph and then Guillaume.

Um, well, I, I just enjoyed today very much.

I thought there was, uh, there was a little bit of both for people from let's say psychology neuroscience, but also from, for engineers.

So if you haven't watched some of the talks, go, go, go look through the schedule because some of the talks were really good.

I think so I really enjoyed that.

Um, and then, yeah, but should I tell people, um, go work out, do, uh, do a lot of sports.

It's good for you.

Mm hmm.

Oh, yeah.

So I was going to make a joke, but I can't because it wouldn't be.

Uh, so I'm going to use my 30 seconds, uh, just to, to thank, uh, Daniel and his team for, for, for this, um, you know, so if you want something to do, um, you should go and watch the live streams, um, and, um, get involved in the podcast.

Get involved with this ecosystem.

I hadn't seen that paper being presented before, but I was really impressed with, you know, with the sort of the active influence Institute and its openness and it's, uh, you know, welcoming, um, um, attitude and vision like the Smithsonian.

Um, so if you want to pursue these ideas, get involved.

Um, and if you, uh, haven't got time, just make sure you attend next year's active influence Institute celebration, but thank you, Daniel.

Rafael?

Yeah.

So, uh, I'll second what, uh, what Carl said and follow on with, uh, it's, it's an invitation, not just, uh, to participate in the, uh, in the active influence Institute, but also the invitations participate in, uh, building this, uh, this Gaia tractor, this, uh, this new way of, of, of doing things that, uh, acknowledges the value of growth.

Uh, and also the value of sustainability joins it all together in this thing called regeneration.

And, uh, it really is, uh, it really is a collective effort is a collective learning effort.

So, um, so, and this also, by the way, also applies to, to the panelists as well.

And I think, uh, obviously what, uh, bird and, uh, Carl are doing, it has immediate, uh, things having to have to do with, with, uh, what we're up or after.

But one of the main things that, that we discussed that we keep discussing is also like this, uh, the inter inter subjectivity and the importance of being able to operate well as humans together.

And that connects directly to cognitive science, psychiatry, uh, and, uh, and, uh, and yeah, so everybody that wants to, to, to be engaged and, uh, and be a part of building a better world should be thinking about how do I, how, what am I doing with my, as an individual or as, uh, as, um.

Employer organization or as a researcher or as a leader, uh, or the family member, what am I doing?

Uh, how does it contribute to, to this, uh, this new, this new non-set non non-equilibrium set of state?

So yeah, that's, that's kind of it.

I probably blew through the 60 seconds, but here it is.

And Guillaume.

